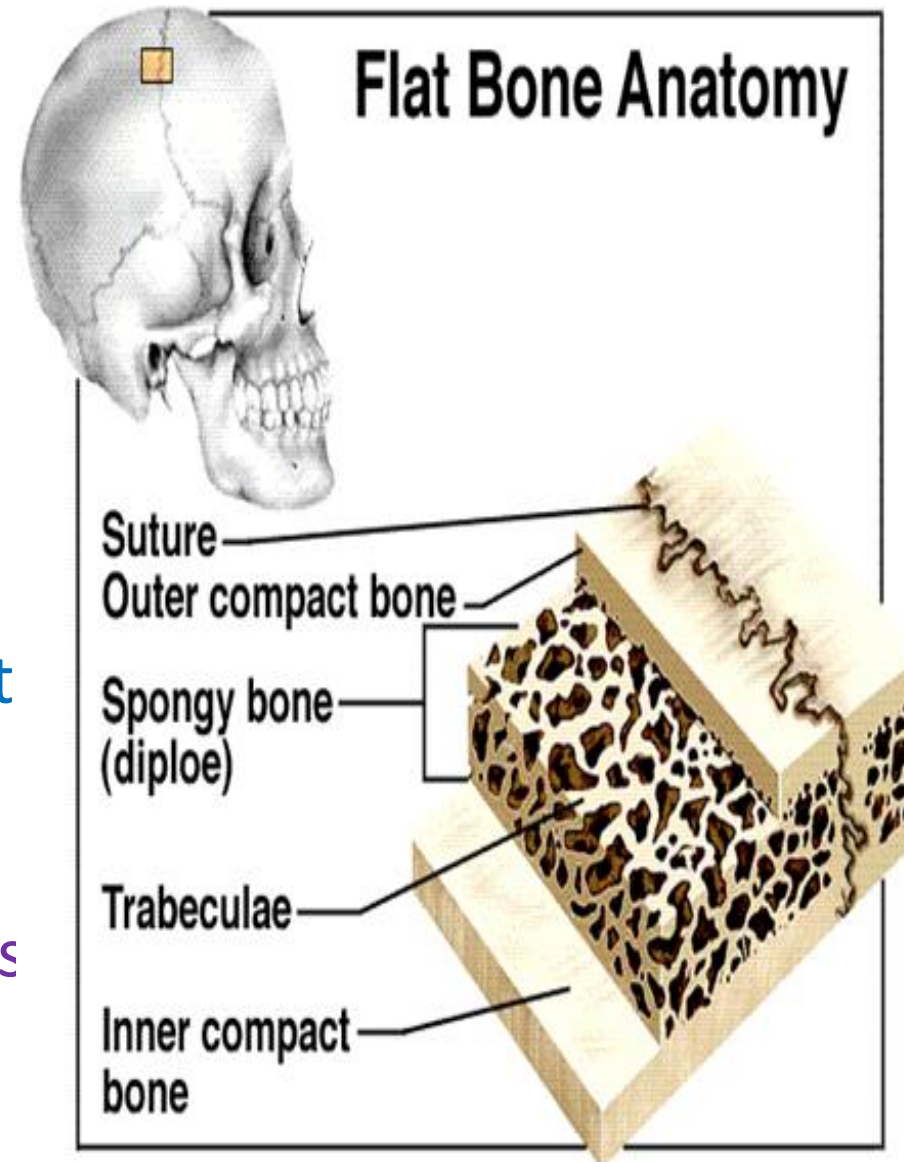


Structure of Short, Irregular, and Flat Bones

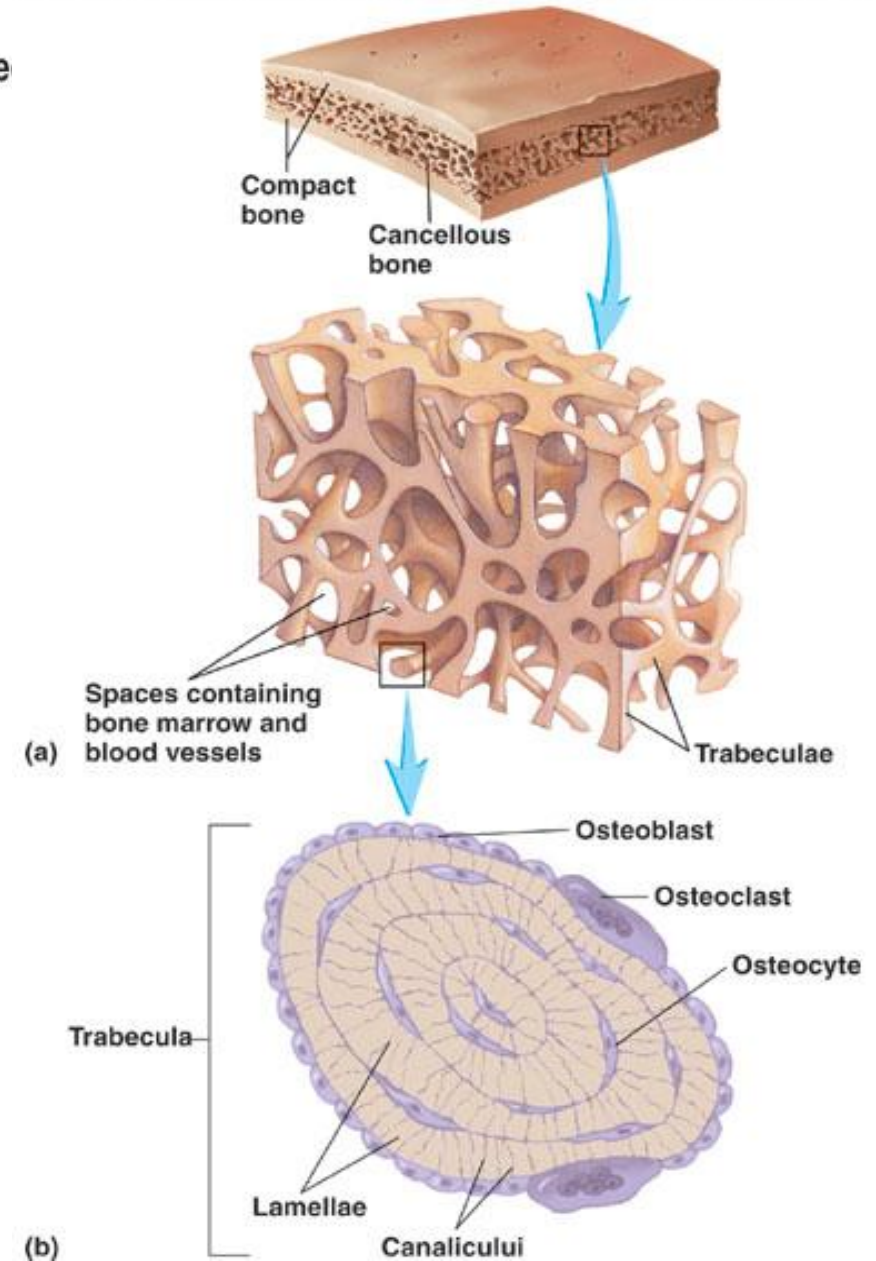
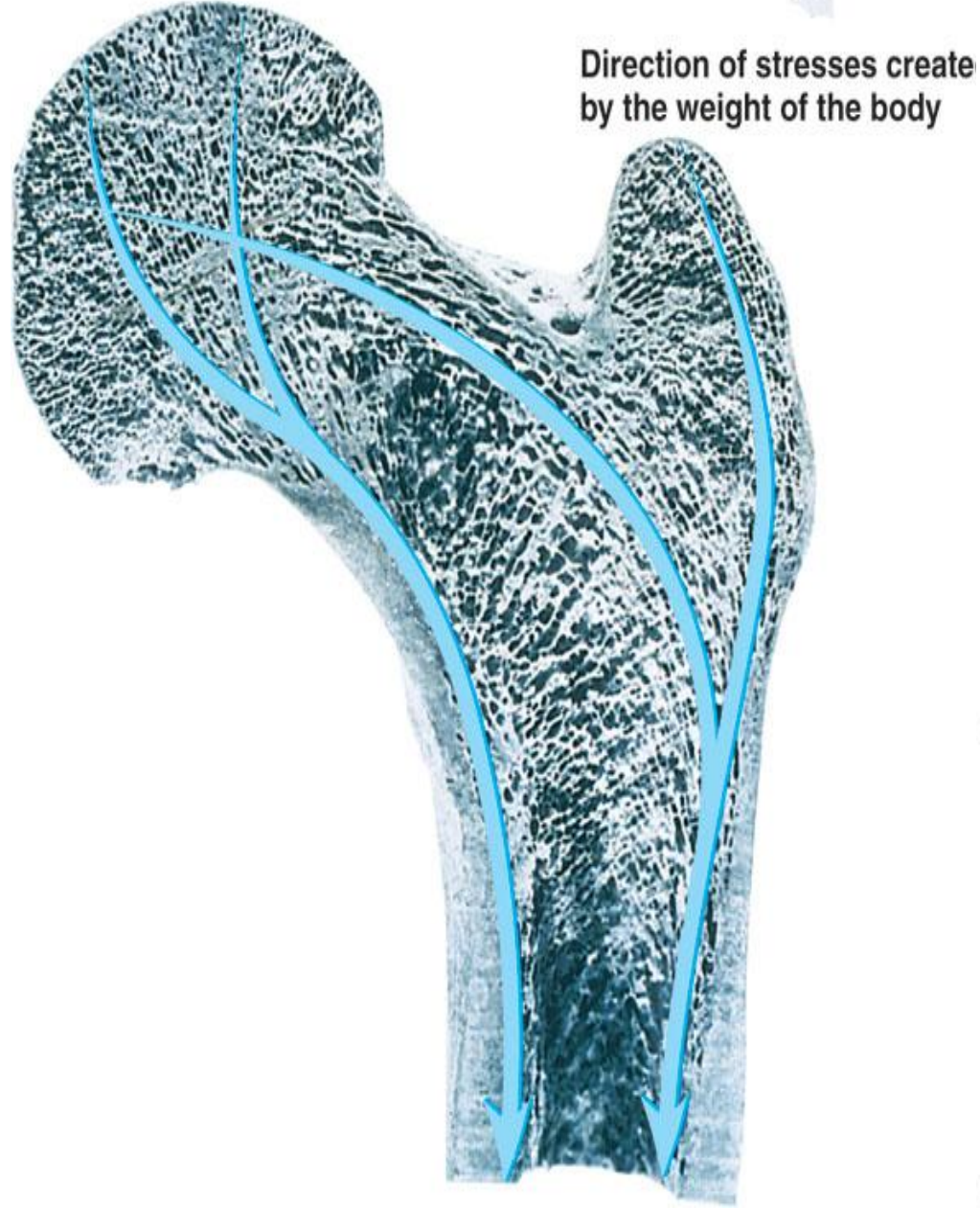
- Thin plates of periosteum-covered compact bone from outside and endosteum-lined spongy bone from inside.
- Have no diaphysis or epiphysis because they are not cylindrical.
- Contain bone marrow between their trabeculae, but no medullary cavity.
- In flat bones, the internal spongy bone layer is known as the **diploë**, and the whole arrangement resembles a stiffened sandwich.



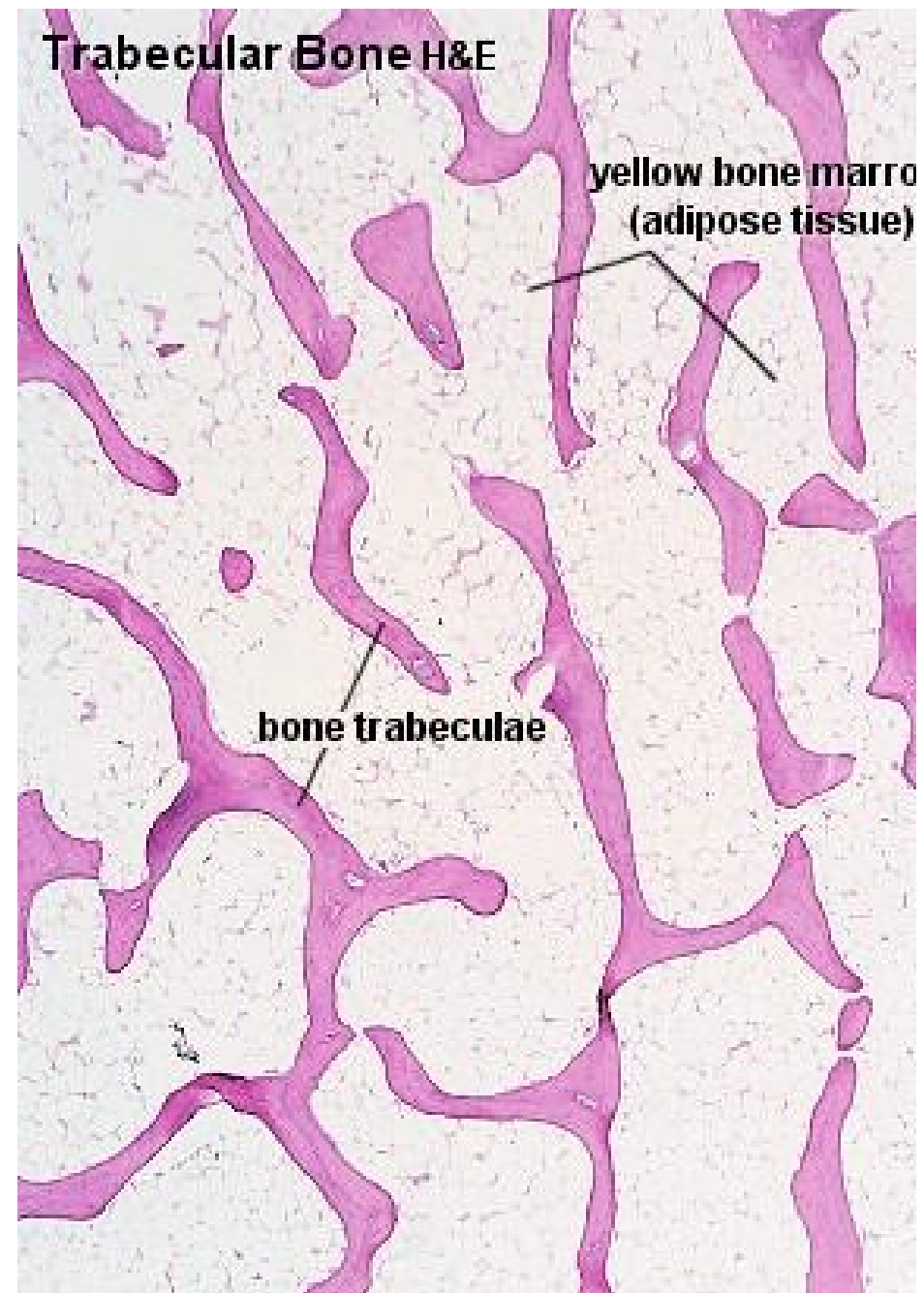
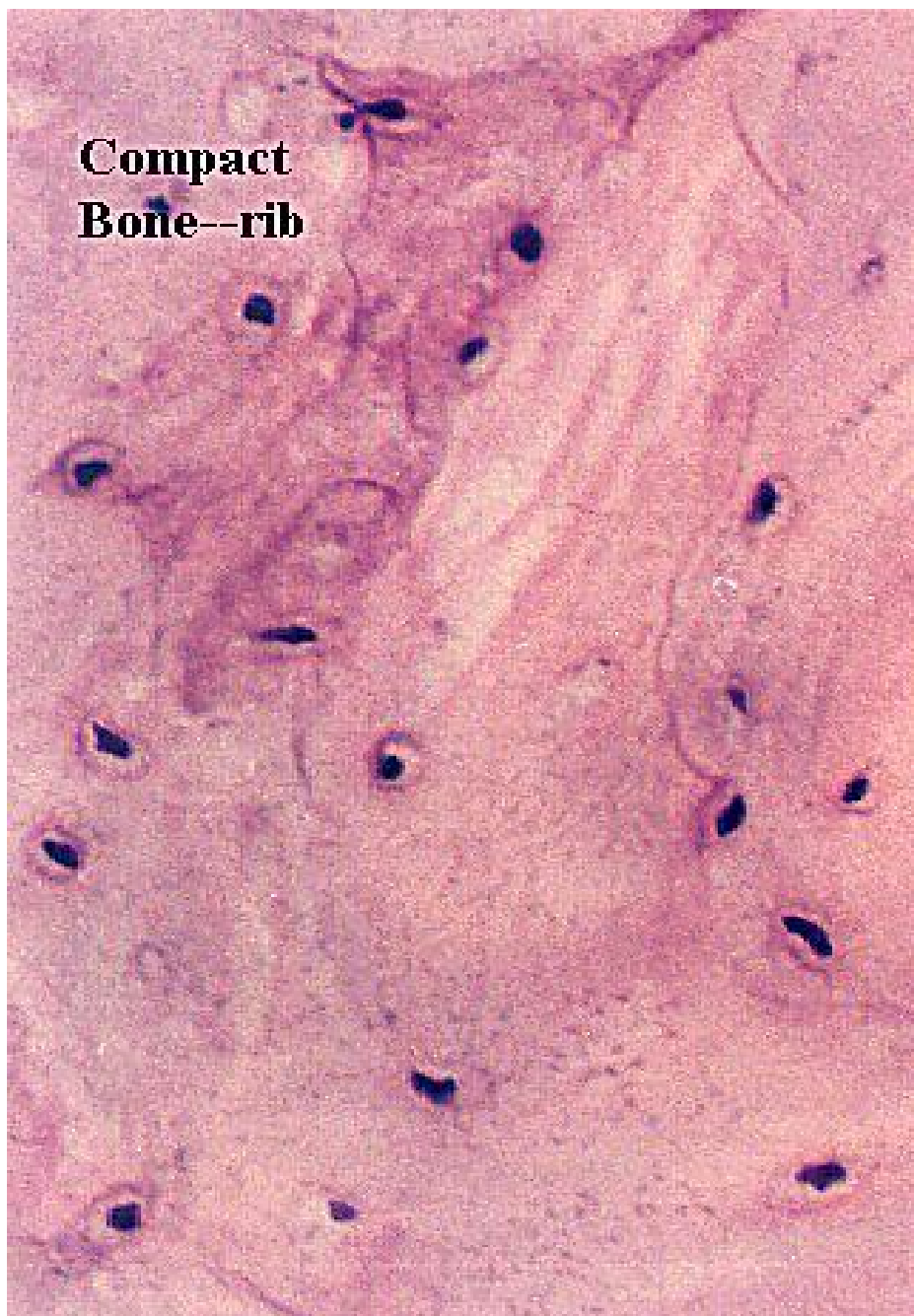
Cancellous (Spongy) Bone

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Compact Bone	Spongy Bone
– Made up of osteons and all other lamellae	– Mainly consists of trabeculae, and trabecular & concentric lamellae
– found beneath periosteum of all bones – makes bulk of diaphyses of long bones	– most of tissue of short, flat, irregular bones – forms epiphyses of long bones & – a rim around medullary cavity of diaphyses of long bones
- dense in texture	- light and open in texture





This bone:

- **Has been demineralized (decalcified in acids)**
- **Has had its inorganic component removed**

BONE MARROW

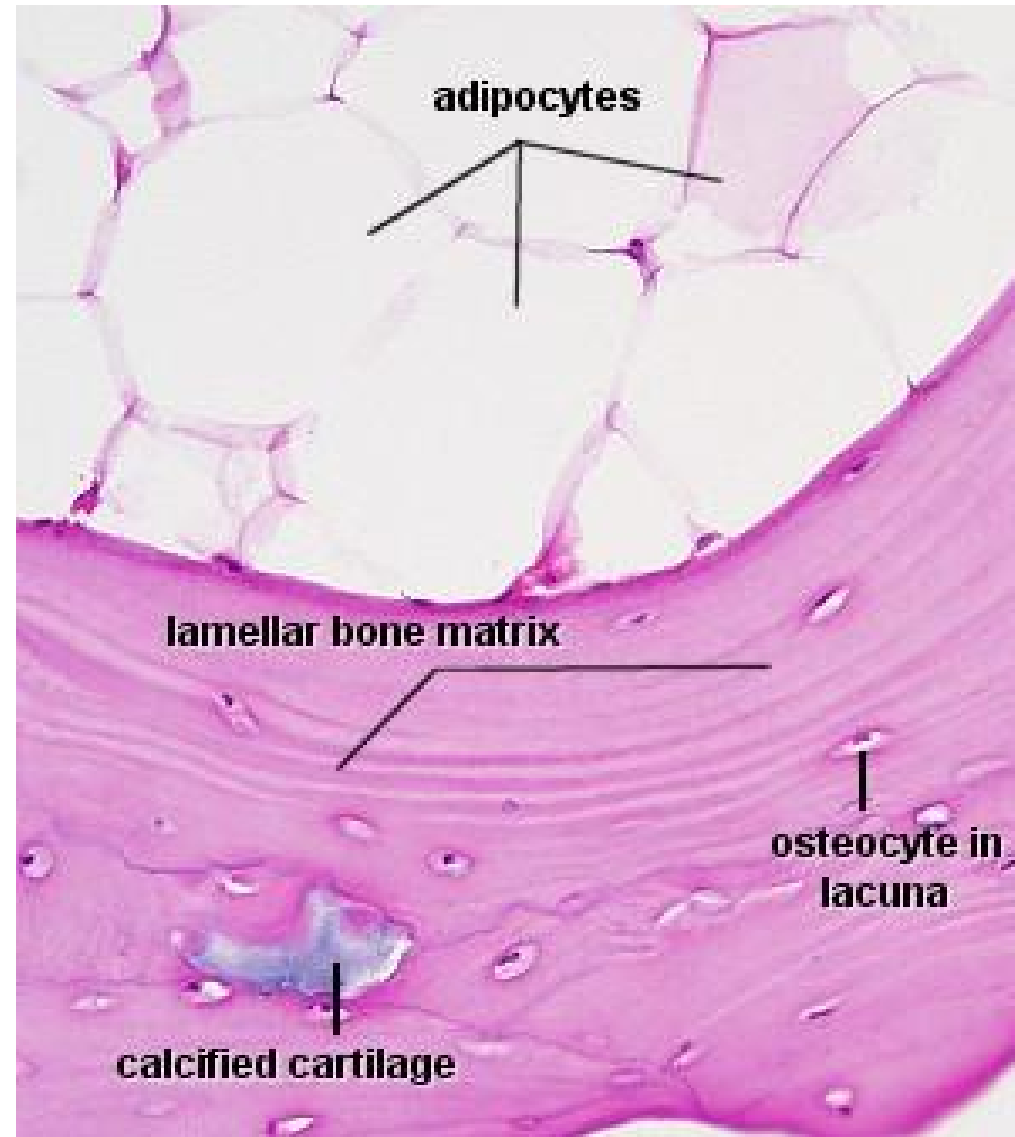
- is a general term for the soft tissue occupying the medullary cavity of a long bone, the spaces b/n the trabeculae of spongy bone, and the larger Haversian canals.
- There are two main types: **red & yellow marrows**.
- **Red bone marrow** = blood cell forming tissue = hematopoietic tissue
 - looks like blood but with a thicker consistency.
 - It consists of a delicate mesh of reticular tissue saturated with immature blood cells and scattered adipocytes.



Distribution of Bone Marrow Cavities

- In a child, the medullary cavity of nearly every bone is filled with **red bone marrow**.
- In young to middle-aged adults, the shafts of the long bones are filled with fatty **yellow bone marrow**.
- Yellow marrow no longer produces blood, although in the event of severe or chronic anemia, it can transform back into red marrow.
- In adults, red marrow is limited to the axial skeleton, pectoral girdle, pelvic girdle, and proximal ends of the humerus and the femur.

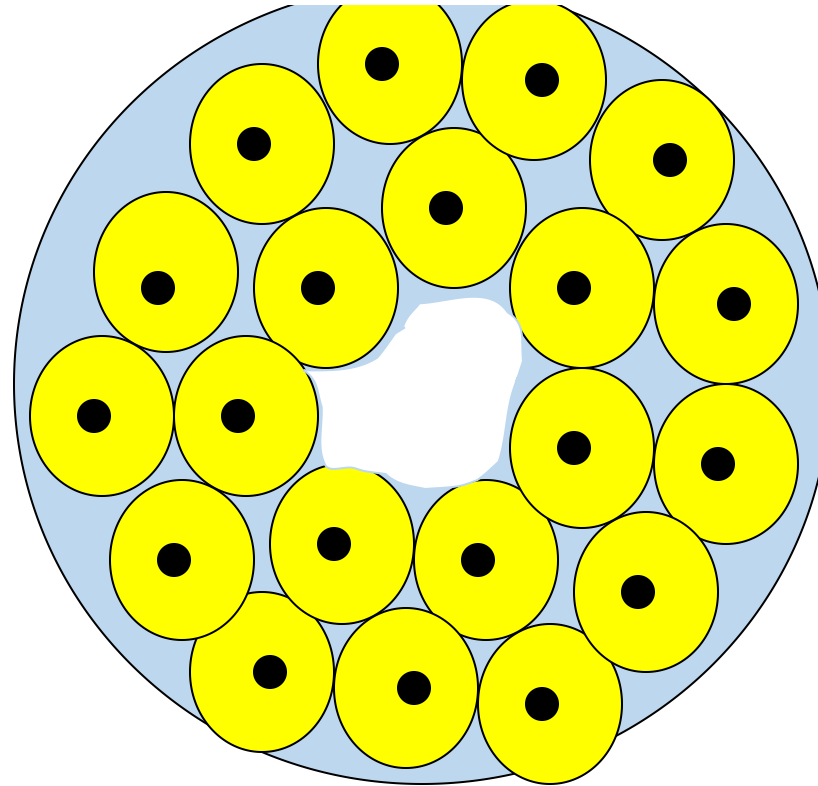
Note the compact bone on the bottom and a yellow marrow on the top.

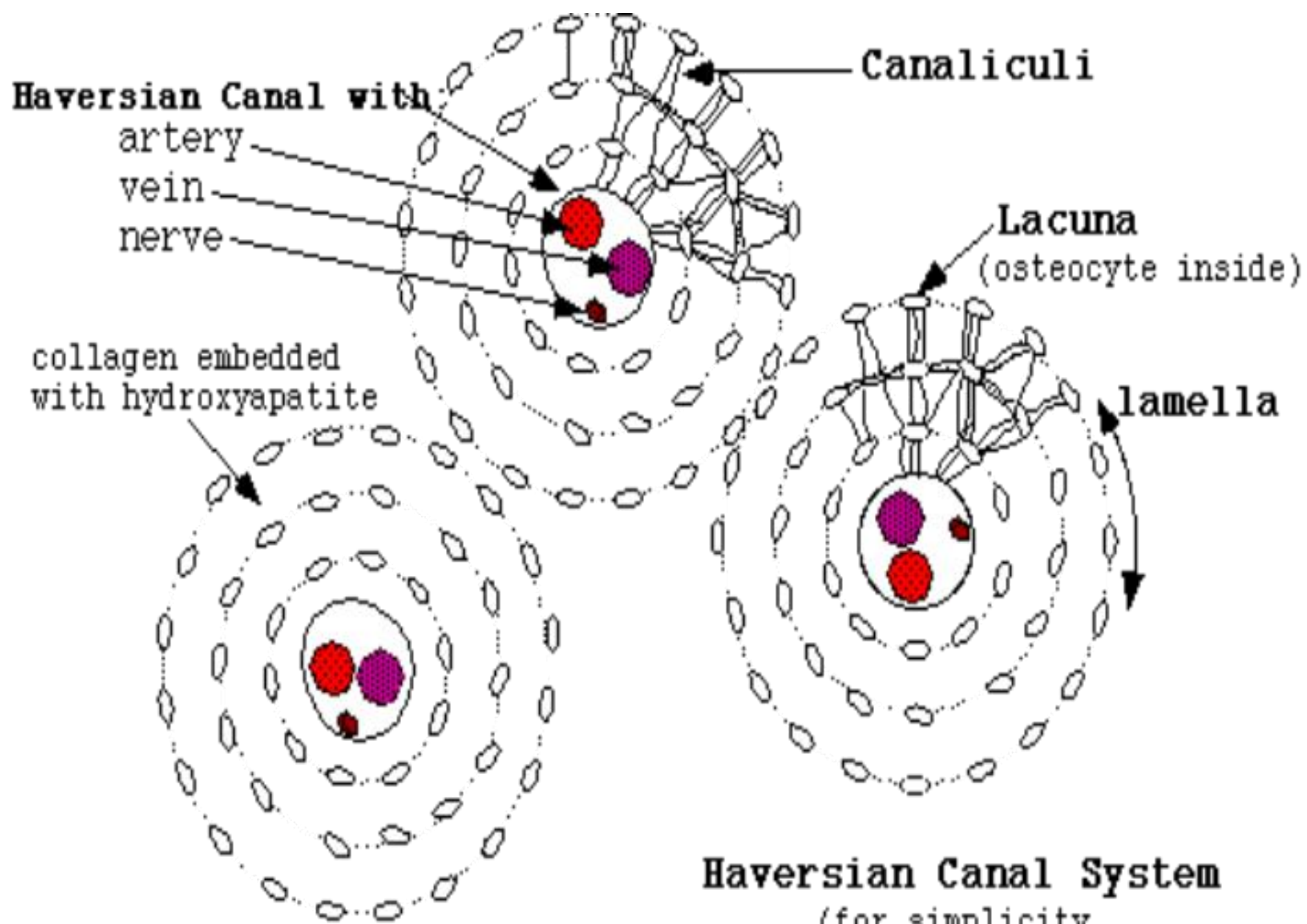


Microscopic Structure of Compact Bone

- Consists of multiple cylindrical structural units known as **osteons** or **Haversian systems**.
- named after a British physician **Clopton Havers**
- Haversian canals are longitudinal canal about 50µm in diameter surrounded by 8-15 lamellae.
- Imagine these osteons as weight-bearing pillars that are arranged parallel to one another along the long axis of a compact bone.

The diagram below represents a long bone shaft in cross-section. **Each yellow circle represents an osteon.** The blue represents bone matrix filling in the spaces b/n osteons. **The white in the middle is the medullary marrow cavity.**





Haversian Canal System
(for simplicity,
only a few canaliculi are shown)

Compact Bone Schmorl stained

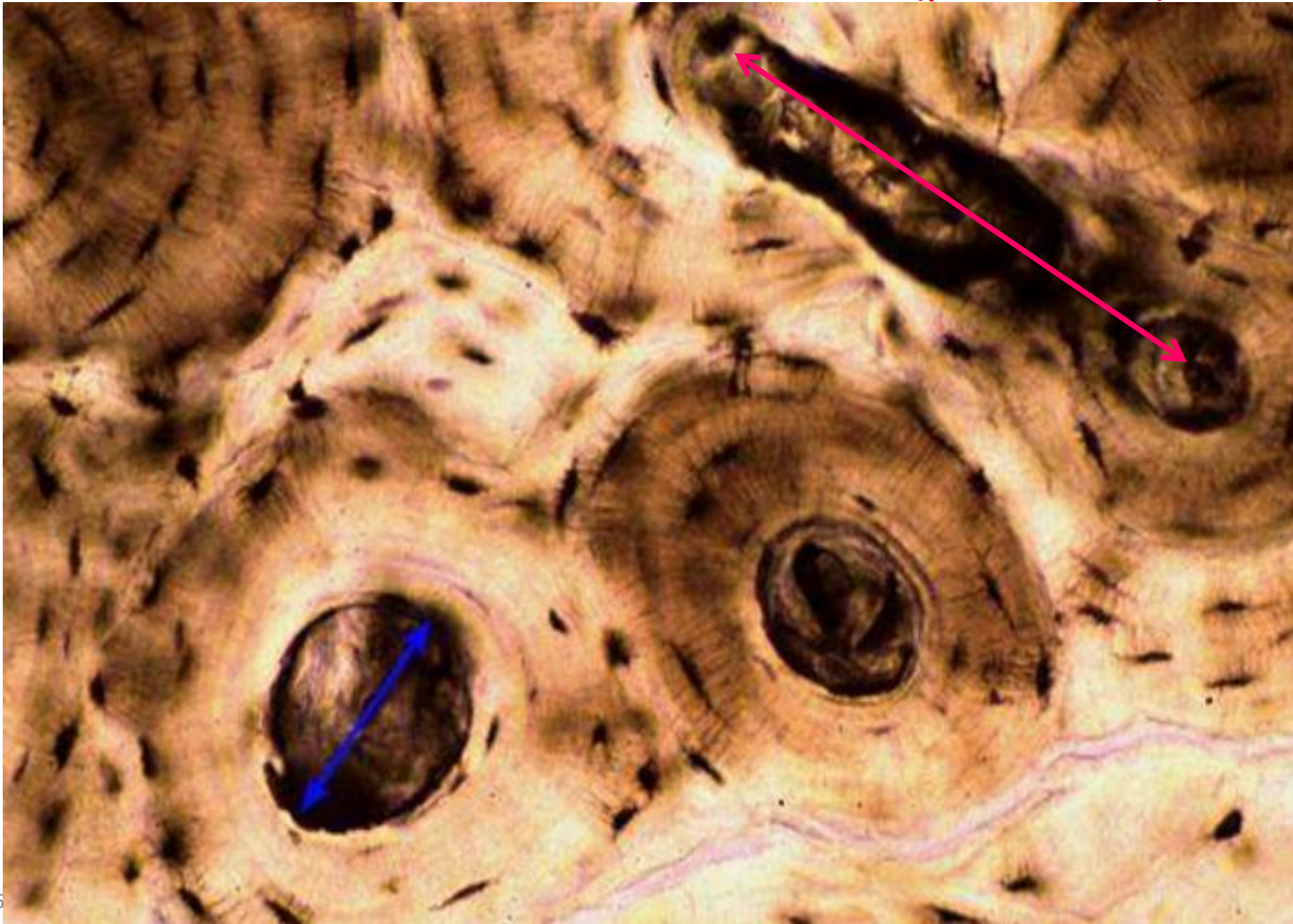
circumferential lamellae

Haversian canal

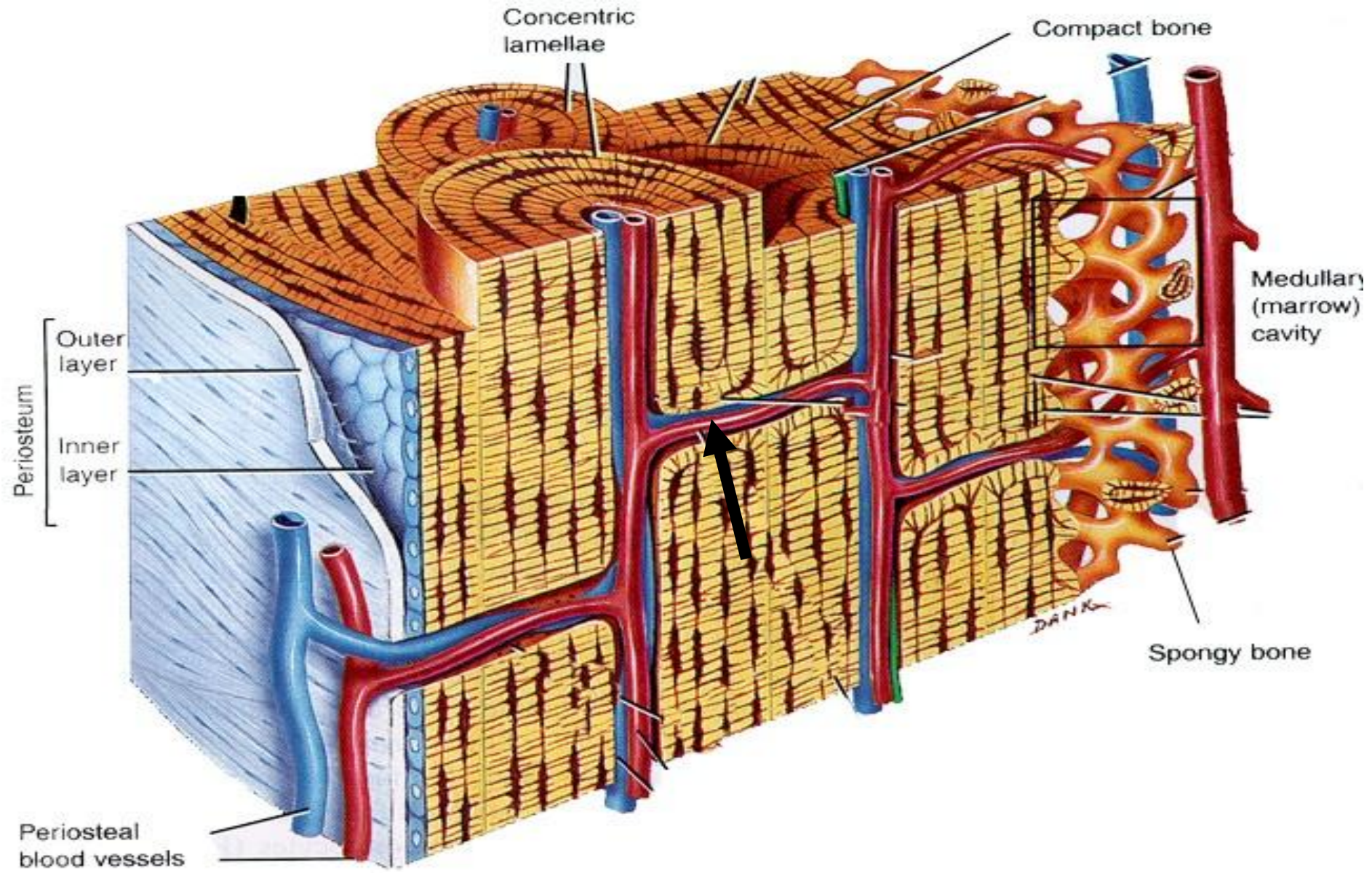
lacuna

Haversian system
(osteon)

Haversian canals and Volkmann's canal (pink arrow)

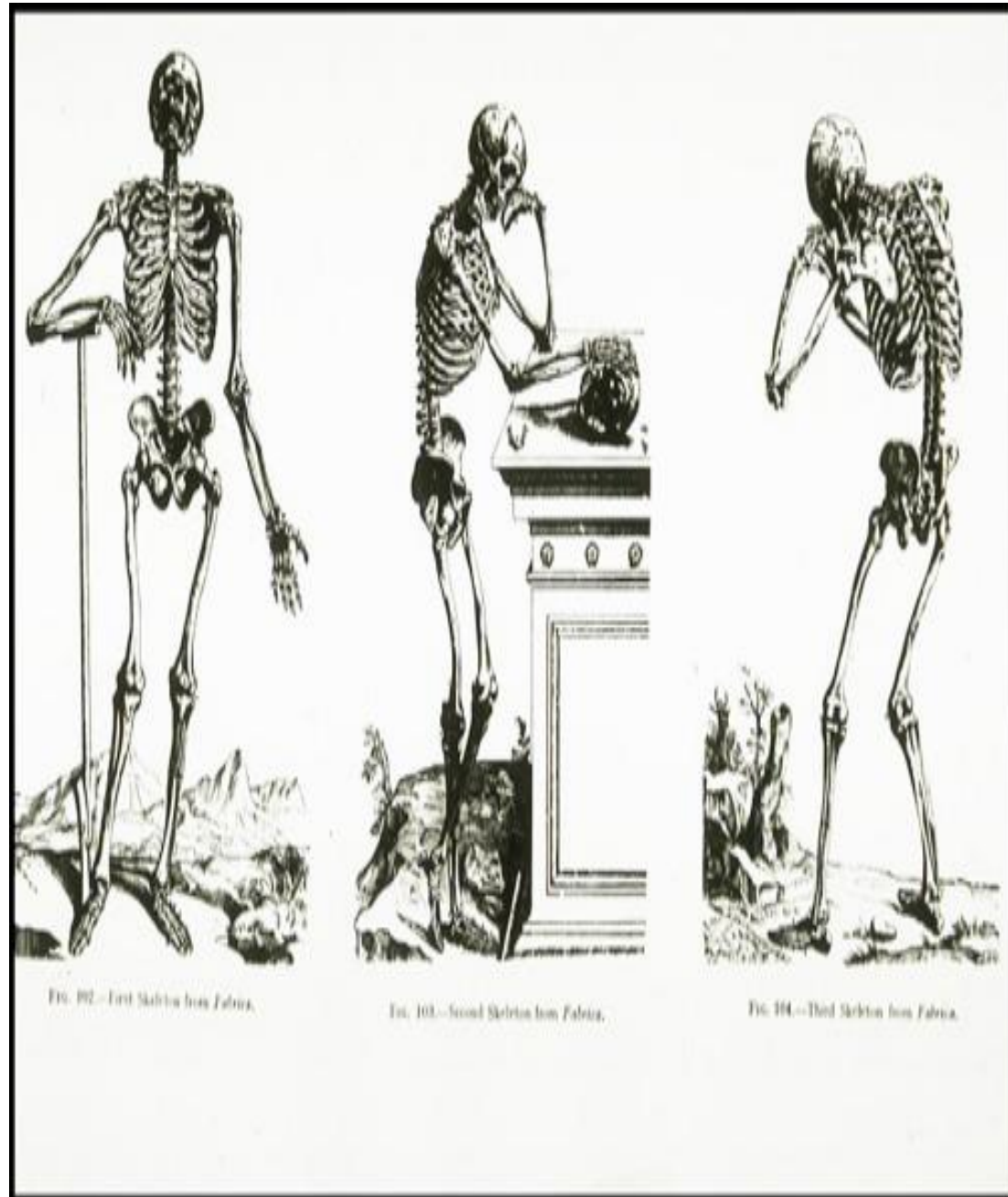


Running perpendicular to the Haversian canals are **Volkman's canals**. They connect the blood and nerve supply in the periosteum to those in the Haversian canals and the medullary cavity.



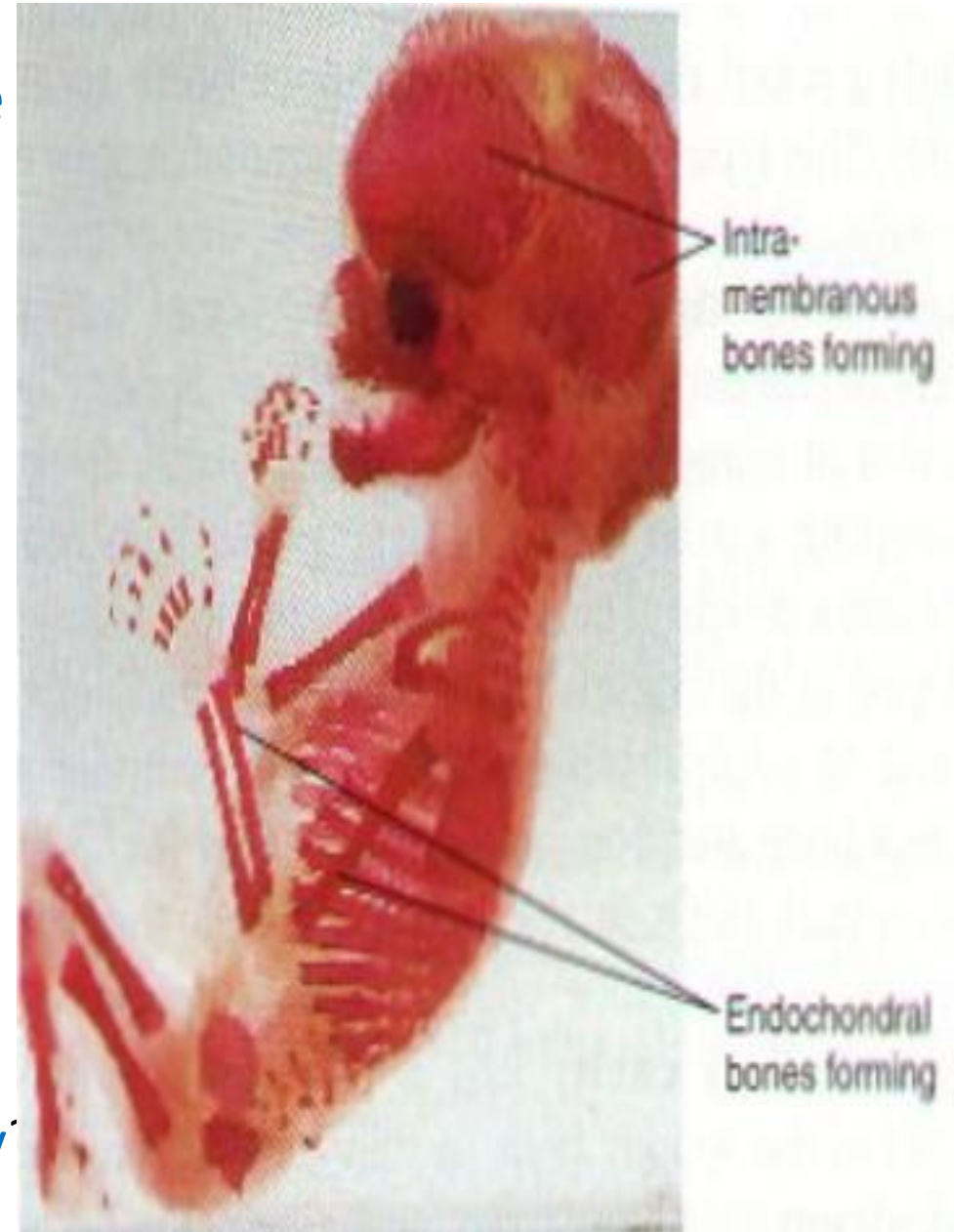
BONE DEVELOPMENT (OSTEOGENESIS)

- (a.k.a. **ossification**) is the process of bone tissue formation.
- In embryos, this leads to the formation of the bony skeleton.
- In children and young adults, ossification occurs as part of bone growth.
- In adults, it occurs as part of bone remodeling and bone repair.

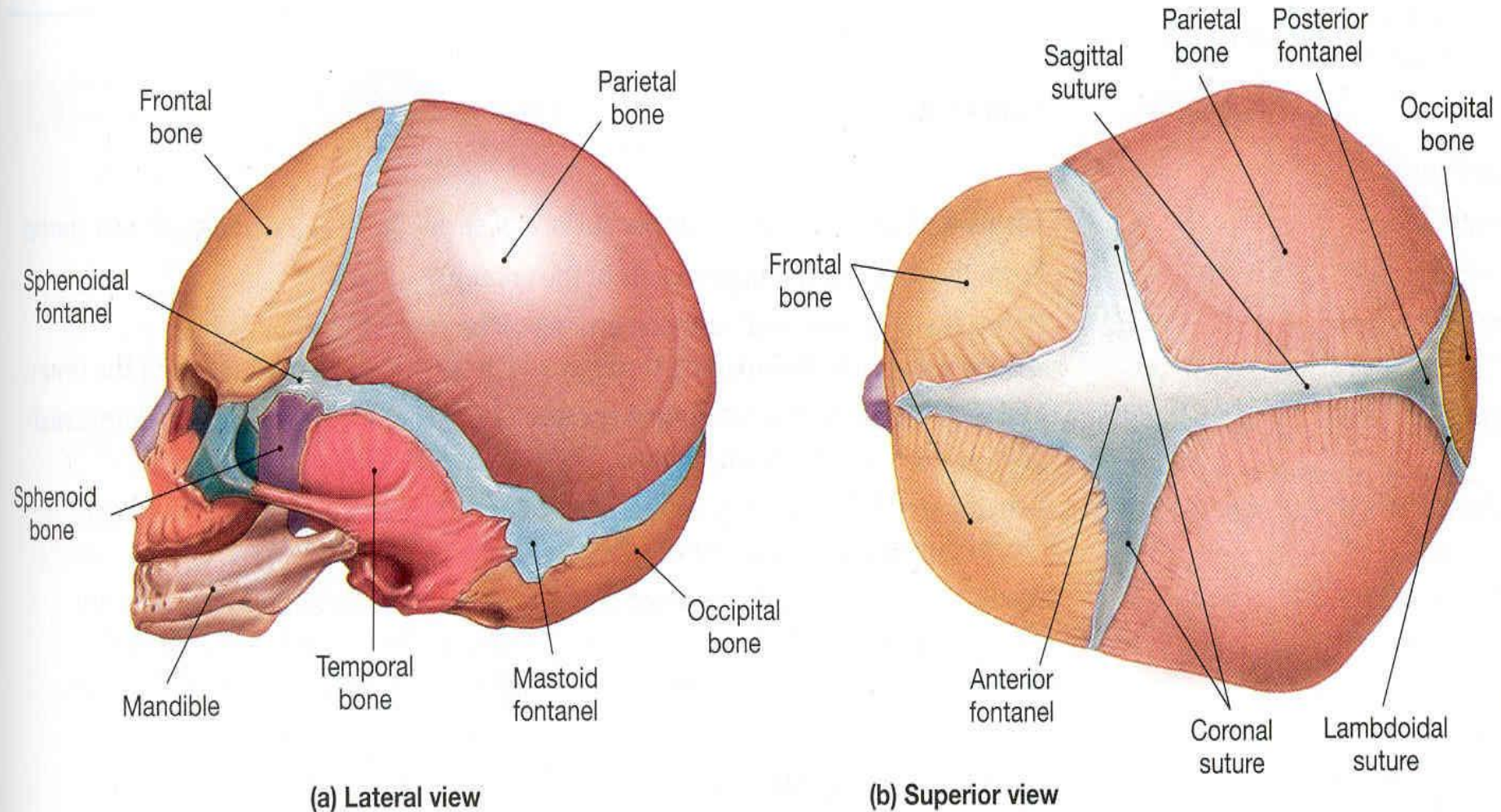


FORMATION OF THE BONY SKELETON

- Before week 8, the human embryonic skeleton is made of fibrous membranes and hyaline cartilages.
- After week 8, bone tissue begins to replace the fibrous membranes and hyaline cartilage.
- The development of bone from a fibrous membrane is called **intramembranous ossification**. Why?
- The replacement of hyaline cartilage with bone is known as **endochondral ossification**. Why?



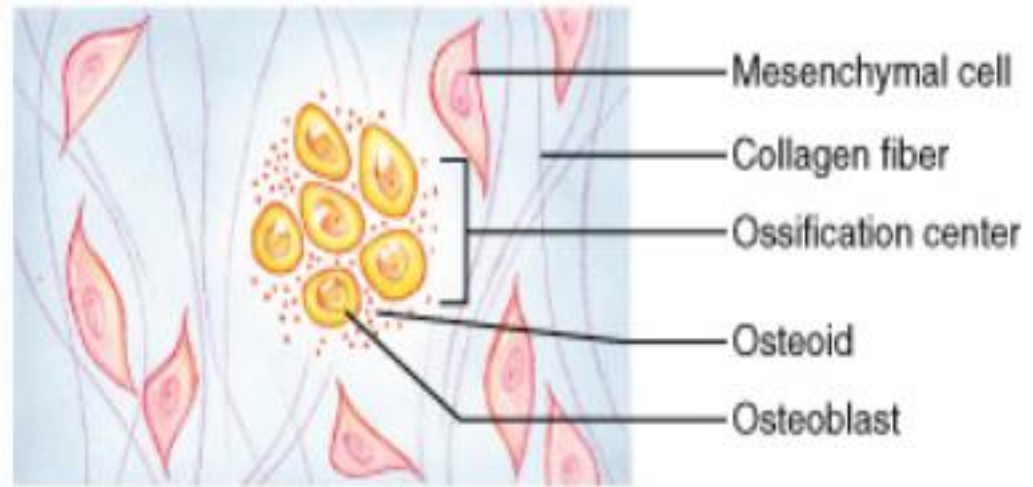
Fontanelles = the fibrous areas between the cranial bones



The Skull of an Infant.

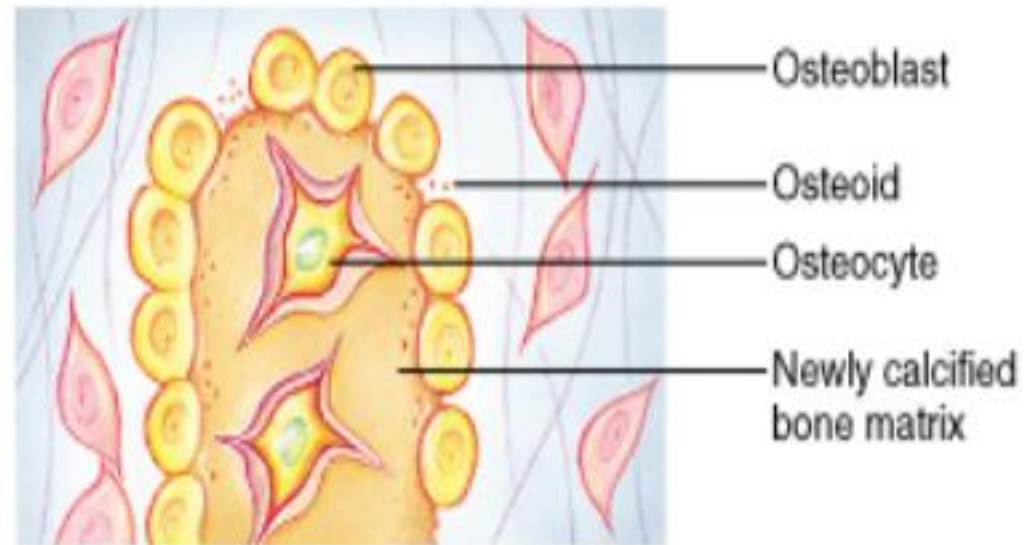
(a) Lateral view. The bones of the infant skull are separated by areas of fibrous connective tissue called fontanelles. (b) Superior view. The large anterior fontanel closes when a child is about 18 months old.

- Mesenchymal cells first retract their process and divide by mitosis to form cluster of bone forming cells, which start to secrete the organic components of bone matrix which then becomes mineralized through the crystallization of calcium salts.

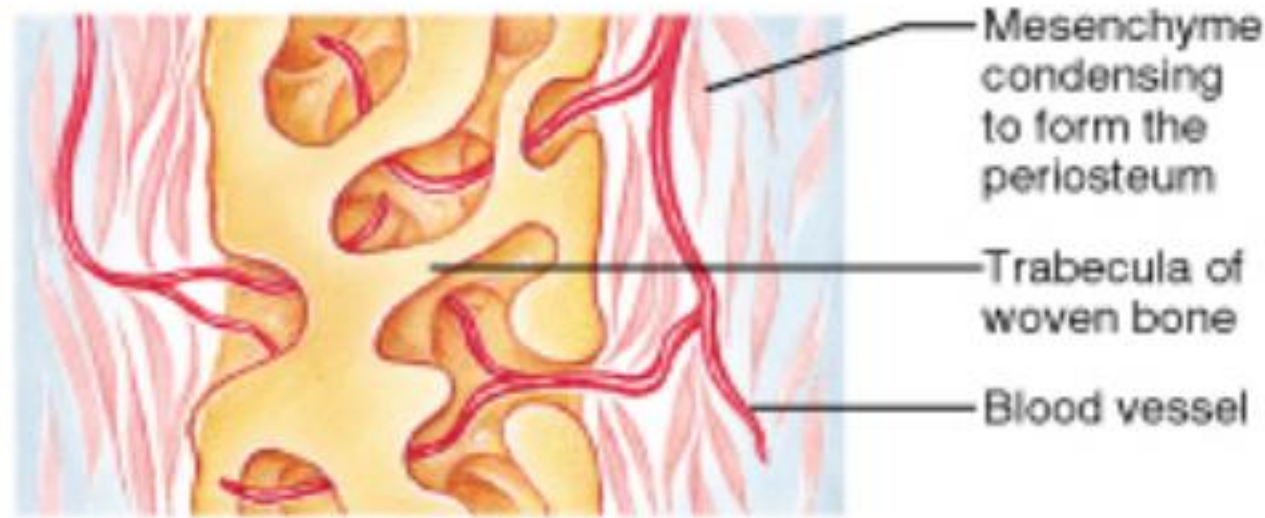


① An ossification center appears in the fibrous connective tissue membrane.

- The location in the tissue where ossification begins is known as an **ossification center**.
- Some osteoblasts are trapped within bony pockets (lacunae).
- These cells differentiate into **osteocytes**.

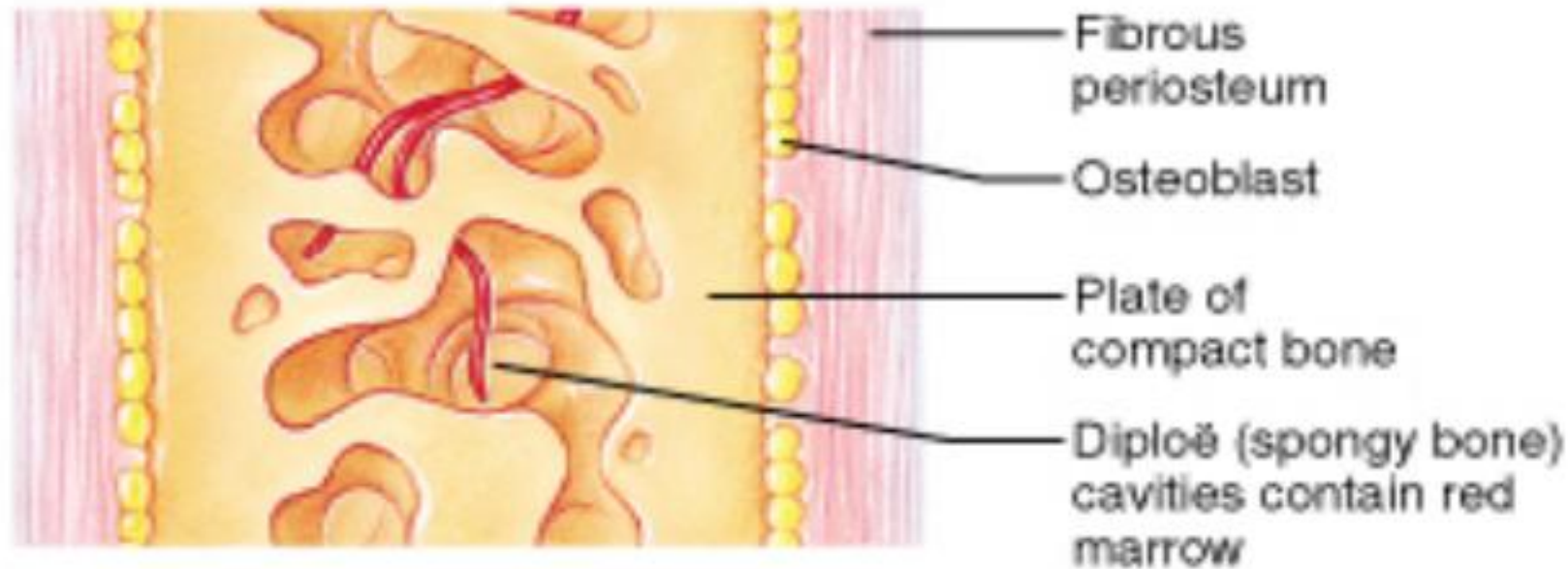


② Bone matrix (osteoid) is secreted within the fibrous membrane.



③ Woven bone and periosteum form.

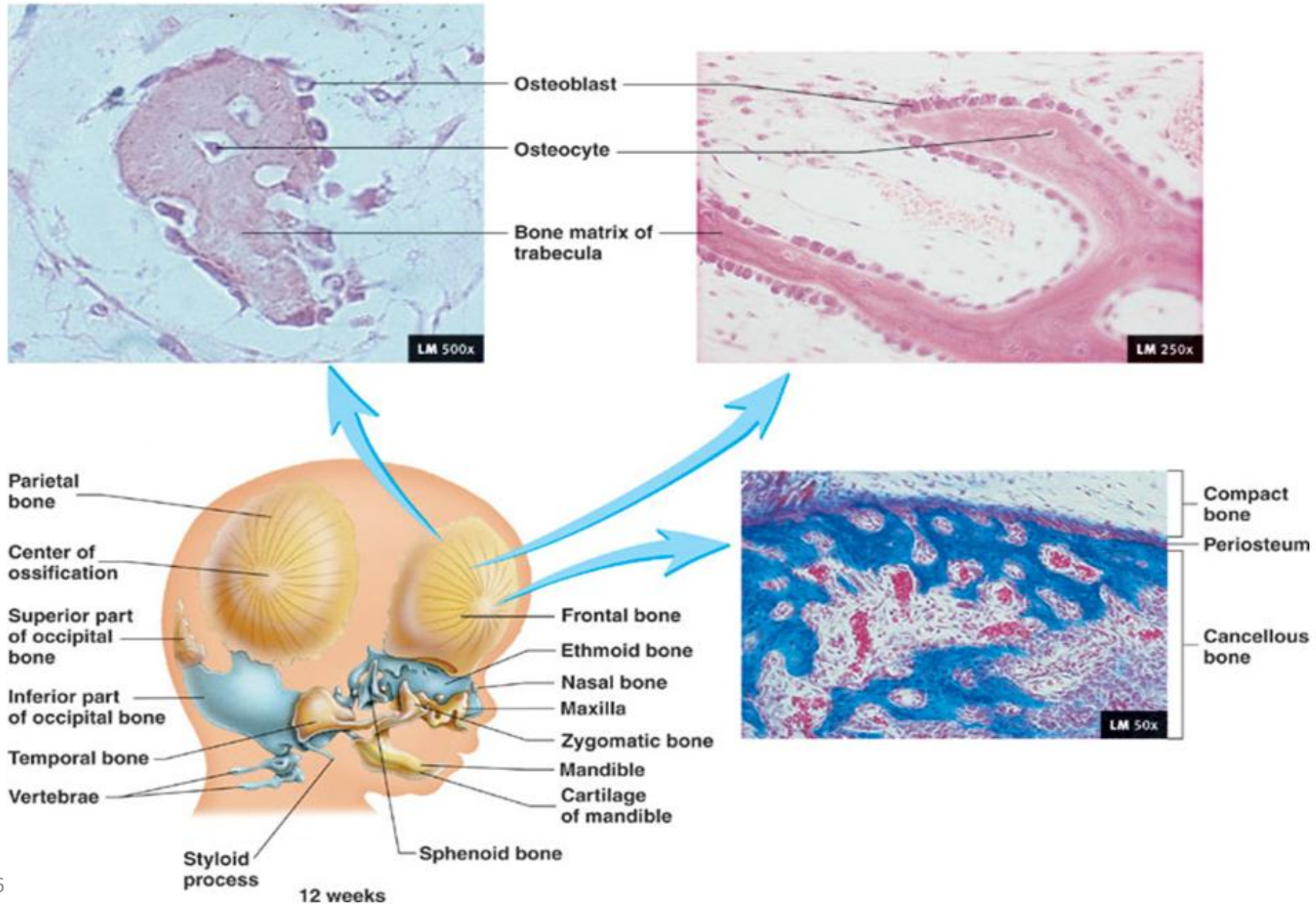
- The developing bone grows outward from the ossification center in small strut called **spicules**.
- **Mesenchymal cell divisions provide additional osteoblasts.**
- The osteoblasts require a reliable source of oxygen and nutrients.
- Blood vessels trapped among the spicules meet these demands and additional vessels branch into the area.
- These vessels will eventually become entrapped within the growing bone.



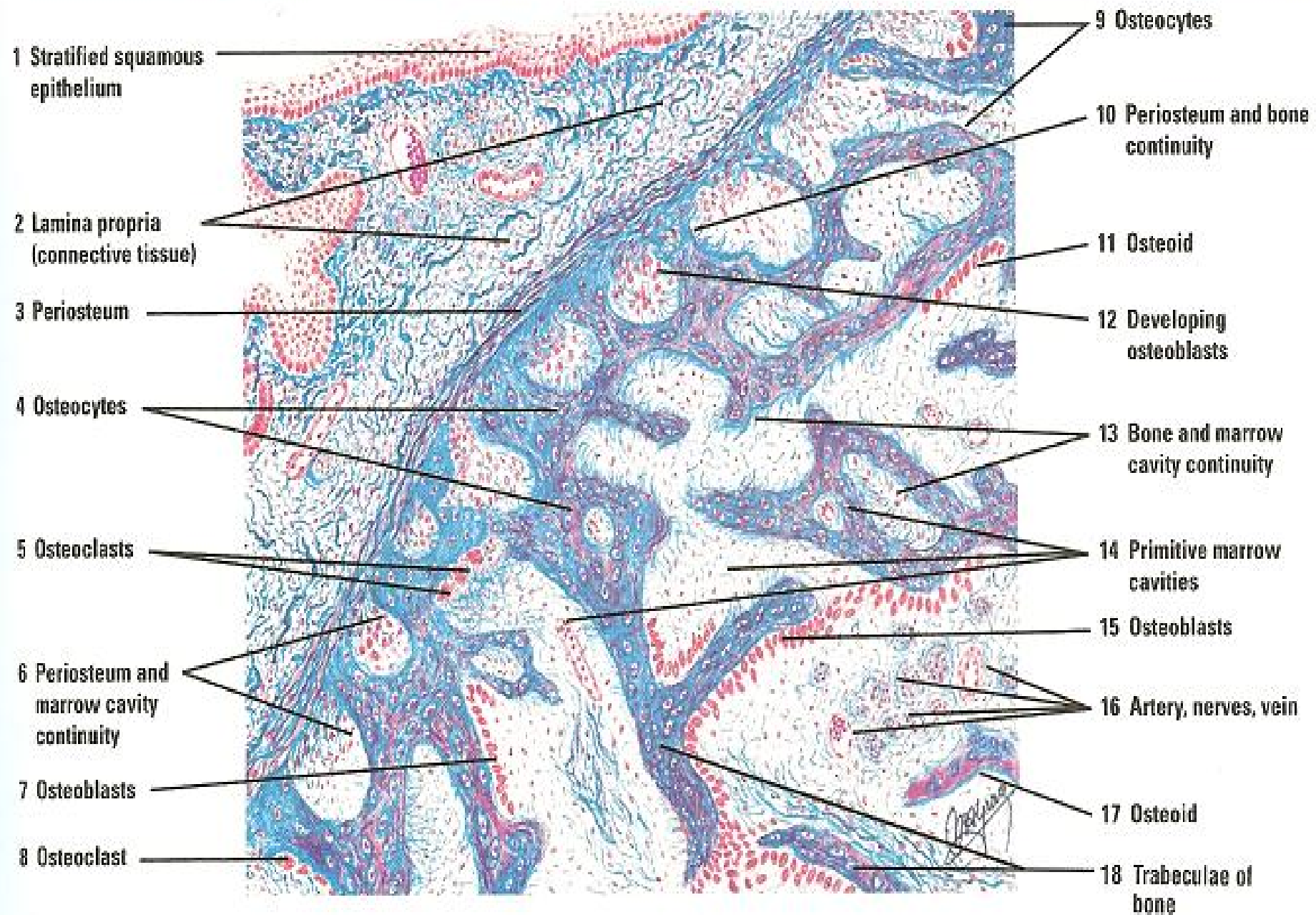
④ Bone collar of compact bone forms and red marrow appears.

- Initially, the intramembranous bone consists only of spongy bone. Subsequent remodeling around trapped blood vessels can produce osteons typical of compact bone.
- As the rate of growth slows, the connective tissue around the bone becomes organized into the fibrous layer of the **periosteum**.
- Osteoblasts close to the bone surface form the inner cellular layer of the periosteum.

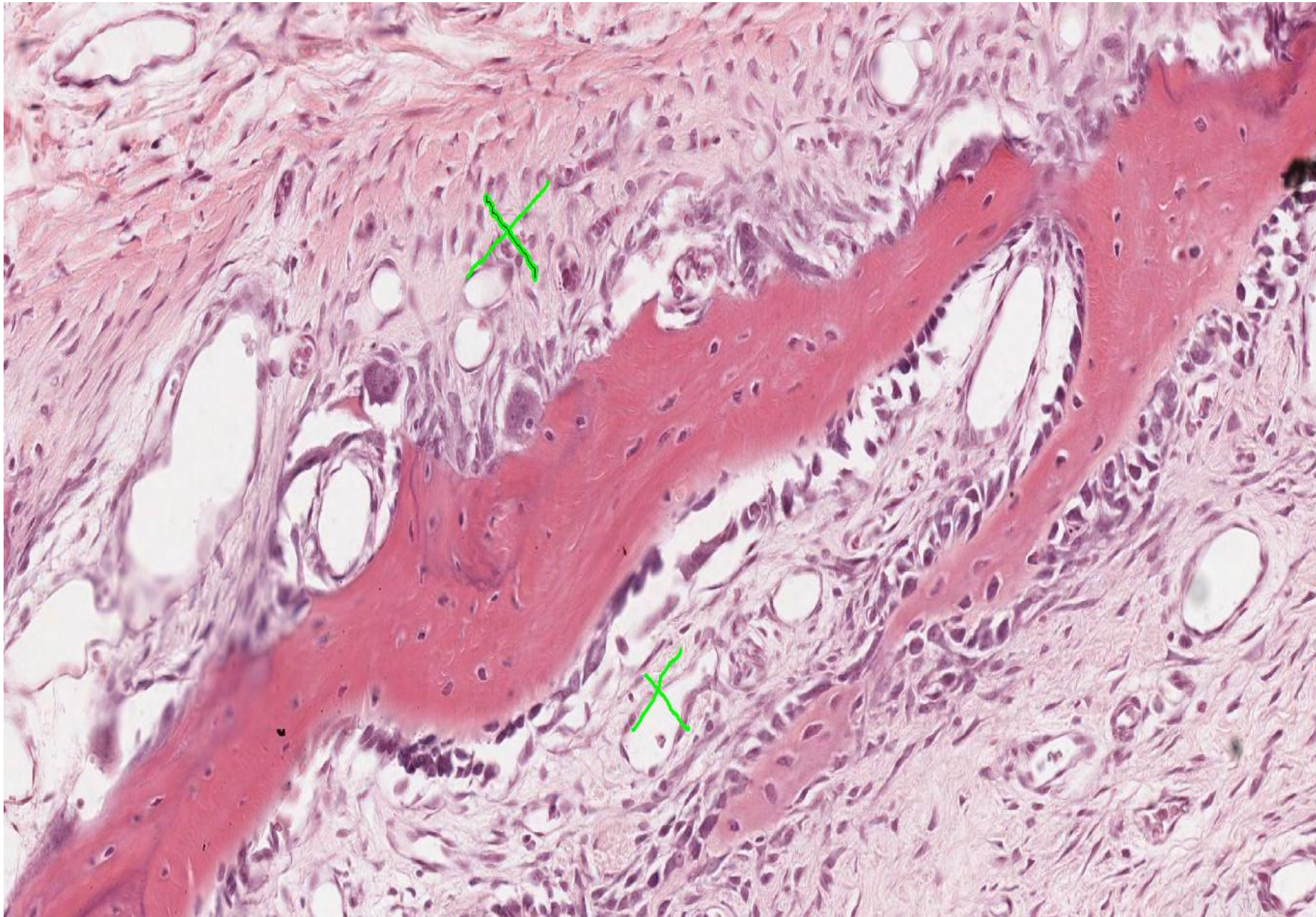
Intramembranous Ossification



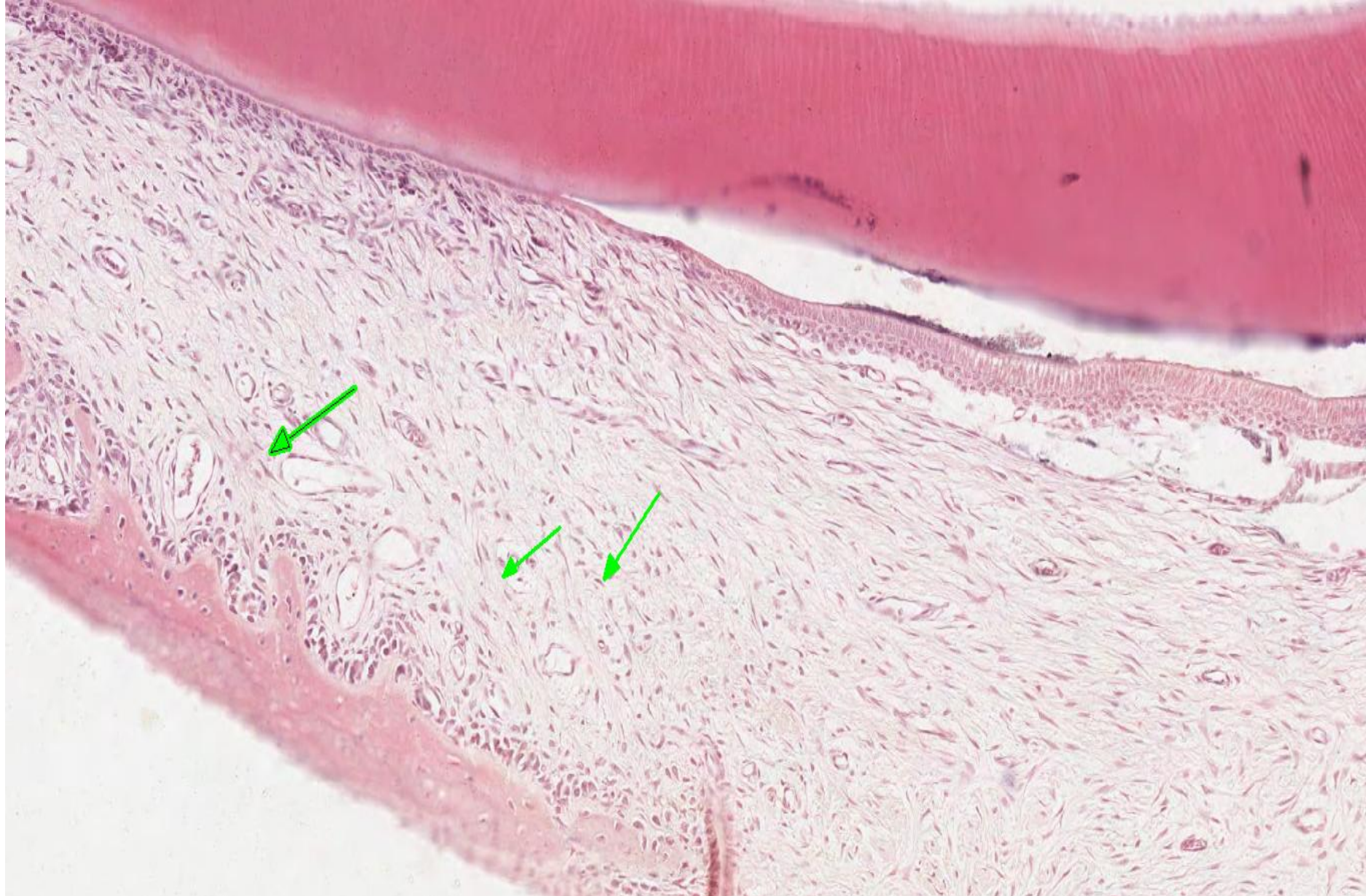
INTRAMEMBRANOUS OSSIFICATION



INTRAMEMBRANEOUS OSSIFICATION



INTRAMEMBRANOUS OSSIFICATION

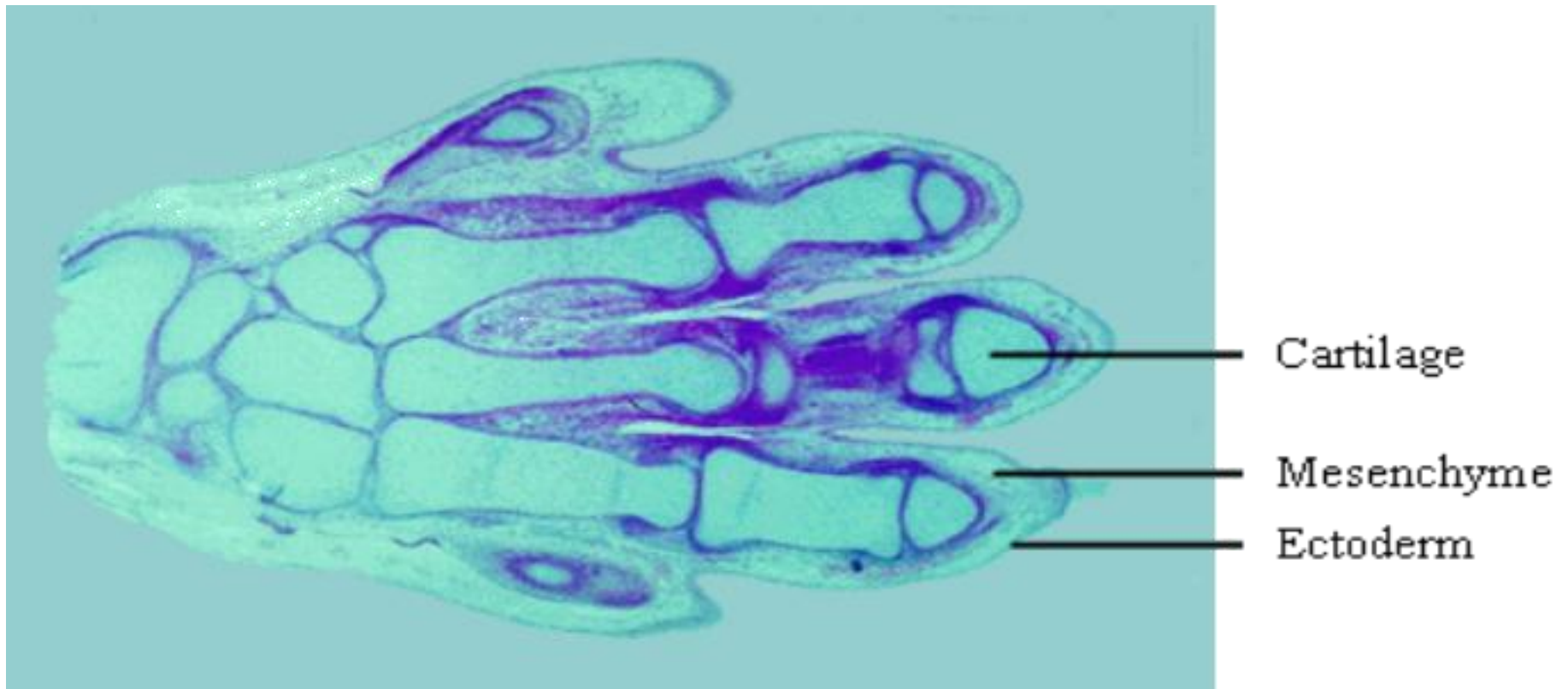


Endochondral Ossification

- Is bone formation from hyaline cartilage models
- Is means of development of bones of the base of the skull, parts of the mandible, epiphyses of the clavicles, all long bones, and most of remaining bones of skeletal system.
- Cartilage formation begins at end of the fourth week of embryonic development.
- Some ossification begins during the third month (weeks 9th -12th); some does not begin until 18-20 years of age.

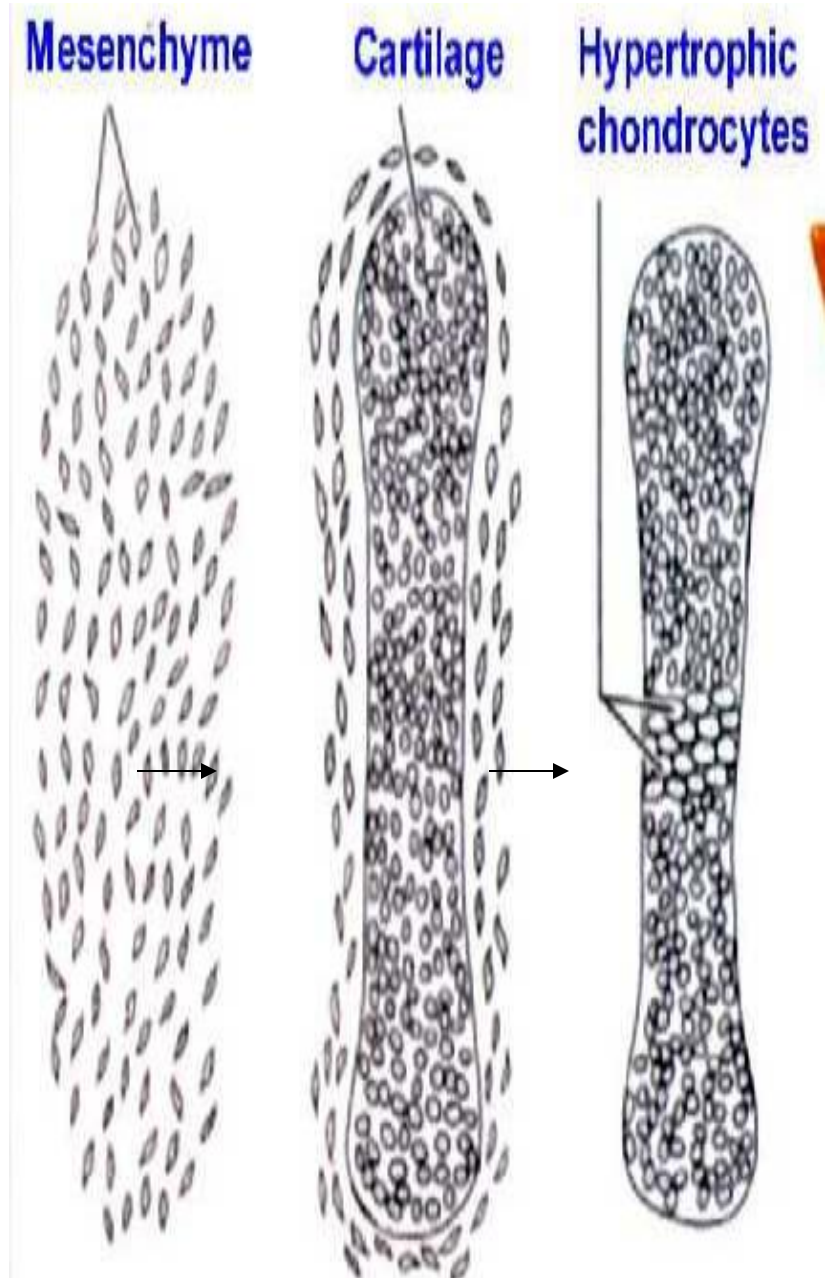
Endochondral Ossification

- Begins with the formation of a hyaline cartilage model which will later be replaced by bone.
- Most bones in the body develop via this model.
- More complicated than intramembranous because the hyaline cartilage must be broken down as ossification proceeds.
- We'll follow limb bone development as an example.



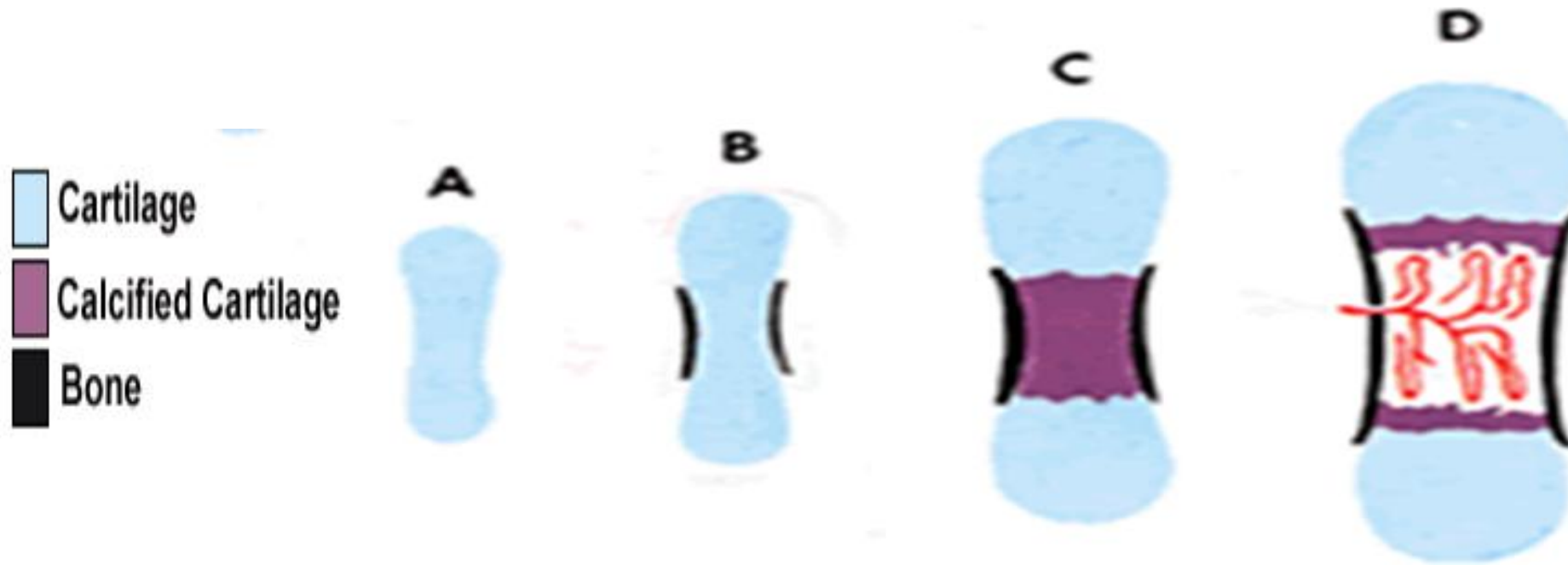
Endochondral Ossification – Step 1

- **Chondrocytes** near the center of the shaft of the hyaline cartilage model increase greatly in size.
- These cells enlarge, their lacunae expand, and the matrix is reduced to a series of thin struts.
- These struts soon begin to calcify.
- The enlarged chondrocytes are now deprived of nutrients (diffusion cannot occur through calcified cartilage) and they soon die and disintegrate.



Endochondral Ossification – Step 2

- Blood vessels grow into the perichondrium surrounding the shaft of the cartilage.
 - The cells of the inner layer of the perichondrium in this region then differentiate into **osteoblasts**.
- The perichondrium is now a periosteum and the inner osteogenic layer soon produces a thin layer of bone around the shaft of the cartilage. This bone is called **perosteal bone collar** and it provides support.



Endochondral Ossification – Step 3

- Blood capillaries from the periosteum and mesenchymal cells migrate into the center of the cartilage, invading the spaces left by the disintegrating chondrocytes.
- The calcified cartilaginous matrix breaks down; the mesenchymal cells differentiate into osteoblasts that replace it with spongy bone.
- Bone development begins at this **primary center of ossification** and spreads toward both ends of the cartilaginous model.
- While the diameter is small, the entire diaphysis is filled with spongy bone.



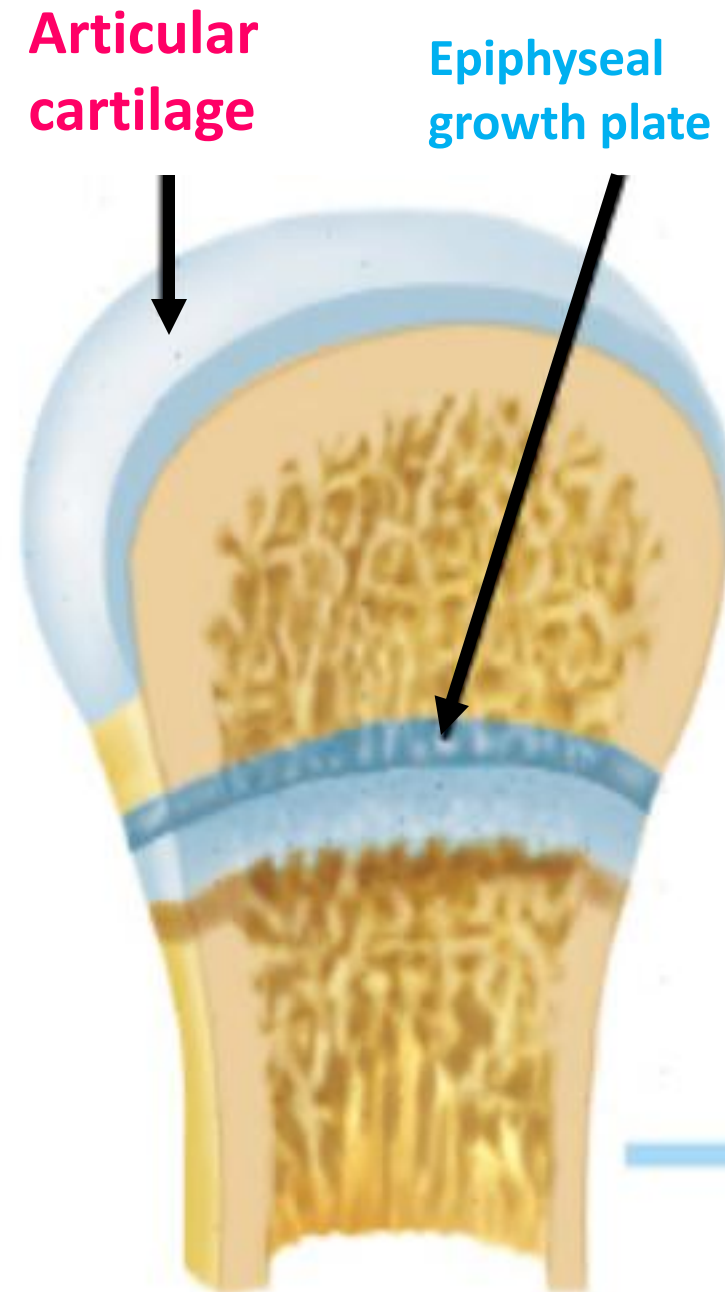
Notice the primary ossification centers in the thigh and forearm bones of the above fetus.

Endochondral Ossification – Step 4

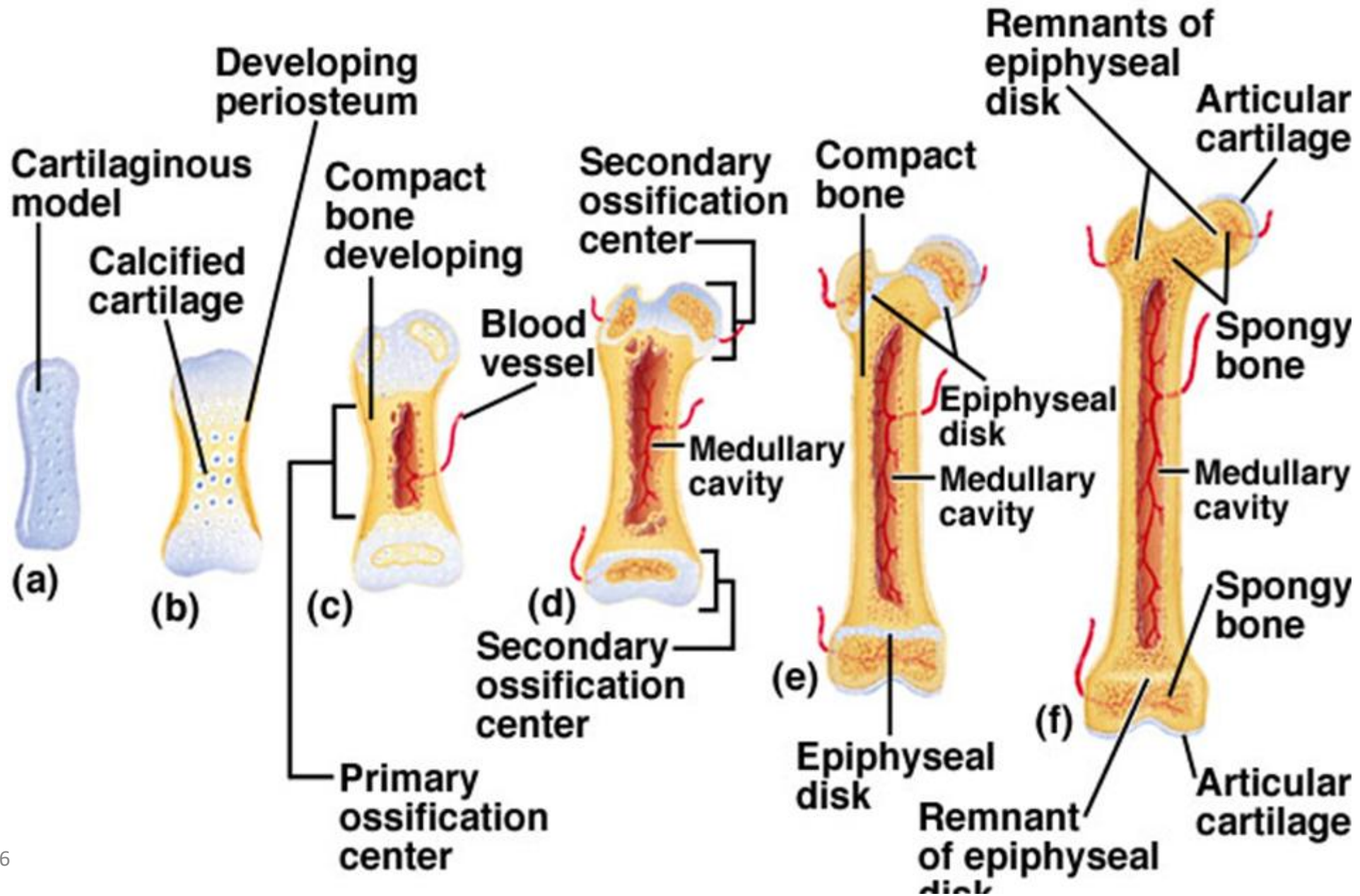
- The primary ossification center enlarges proximally and distally, while osteoclasts break down the newly formed spongy bone and open up a medullary cavity in the center of the diaphysis.
- As the osteoblasts move towards the epiphyses, the epiphyseal cartilage is growing as well.
- *Thus, even though the shaft is getting longer, the epiphyses have yet to be transformed into bone.*

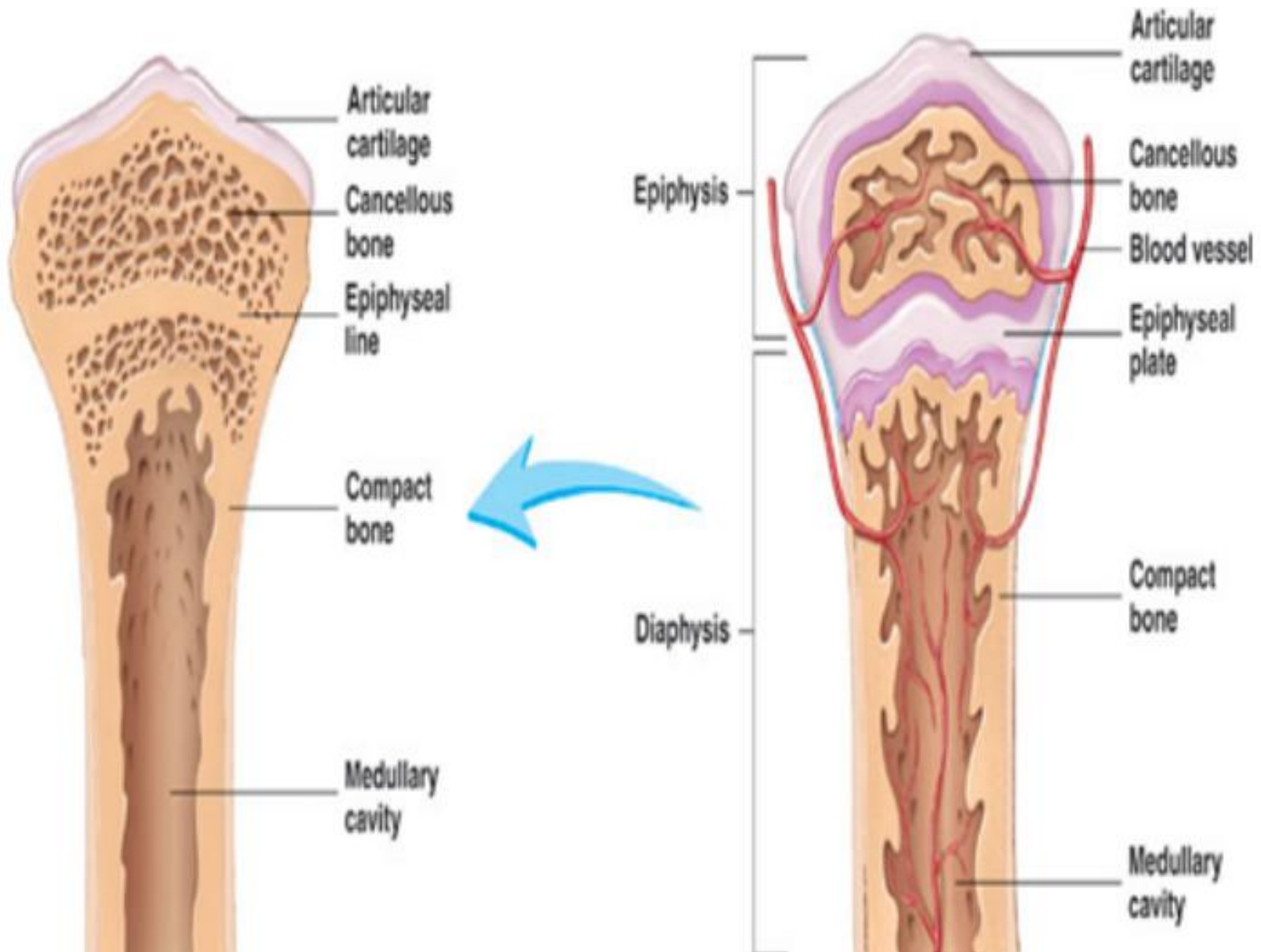
Endochondral Ossification – Step 5

- Around birth, most long bones have a bony diaphysis surrounding remnants of spongy bone, a widening medullary cavity, and two cartilaginous epiphyses.
- At this time, capillaries and osteoblasts will migrate into the epiphyses and create **secondary ossification centers**.
- The epiphysis will be transformed into spongy bone.
- However, a small cartilaginous plate, known as the **epiphyseal growth plate/disc**, will remain at the juncture b/n the epiphysis and the diaphysis.

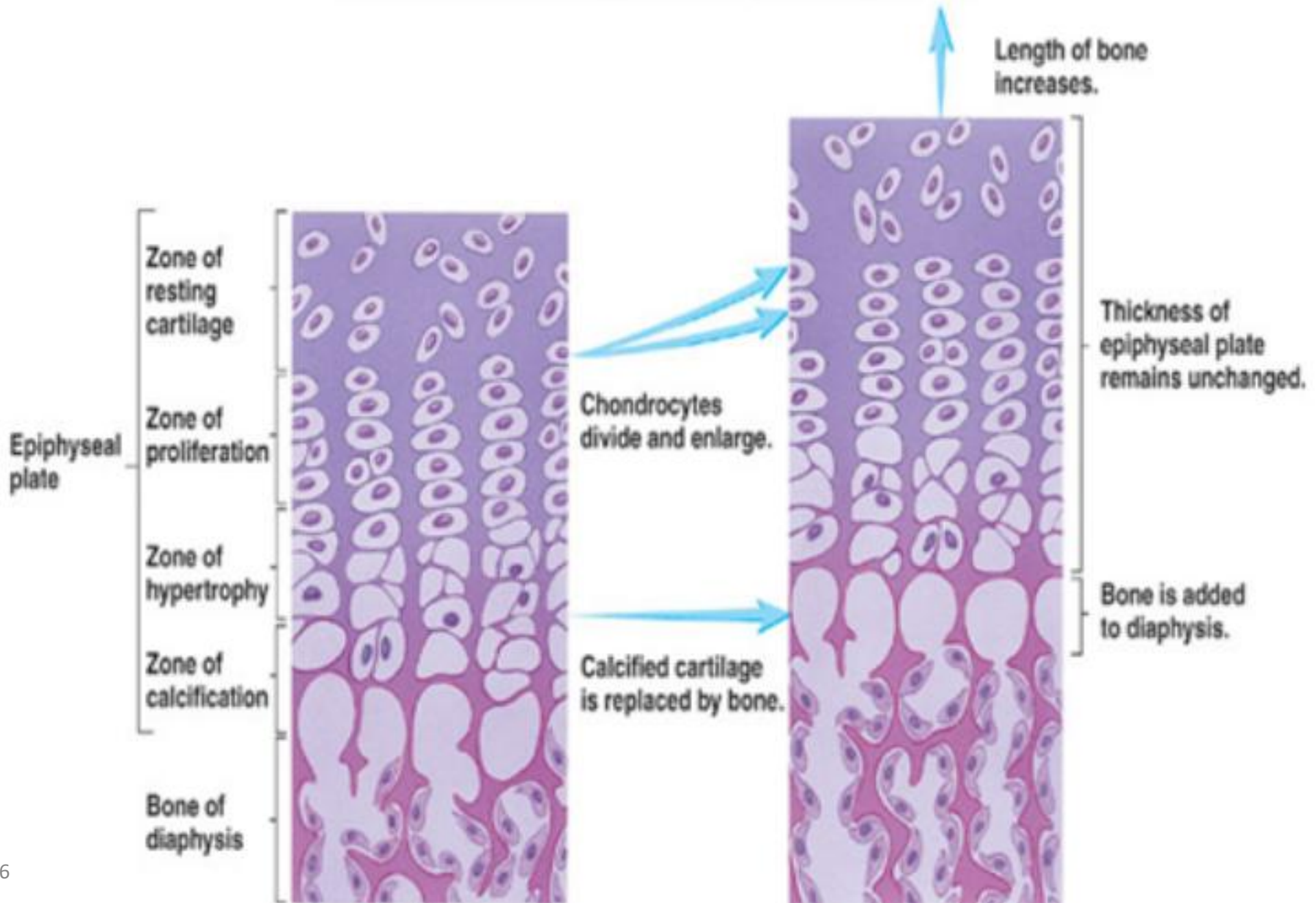


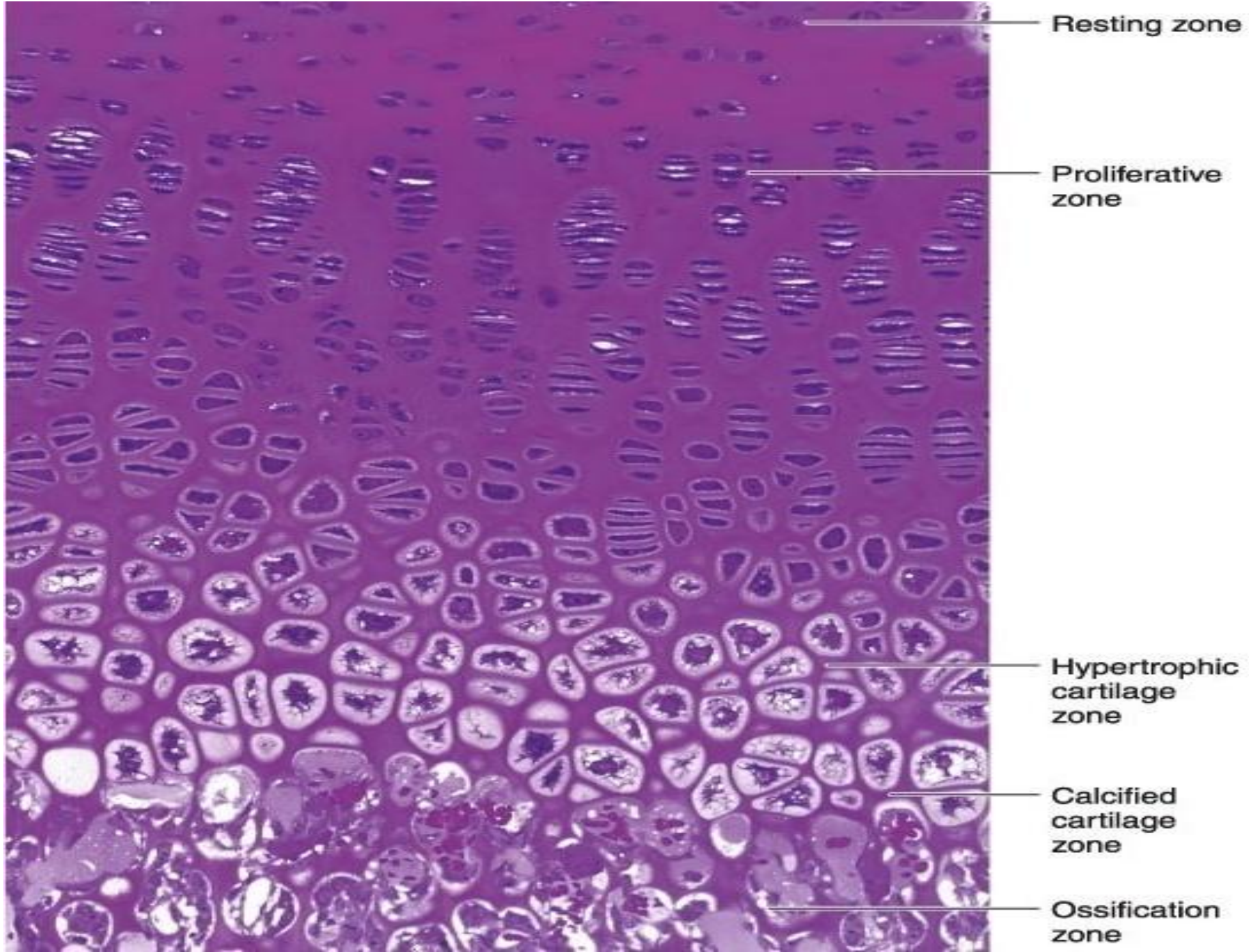
Endochondral Bone Development



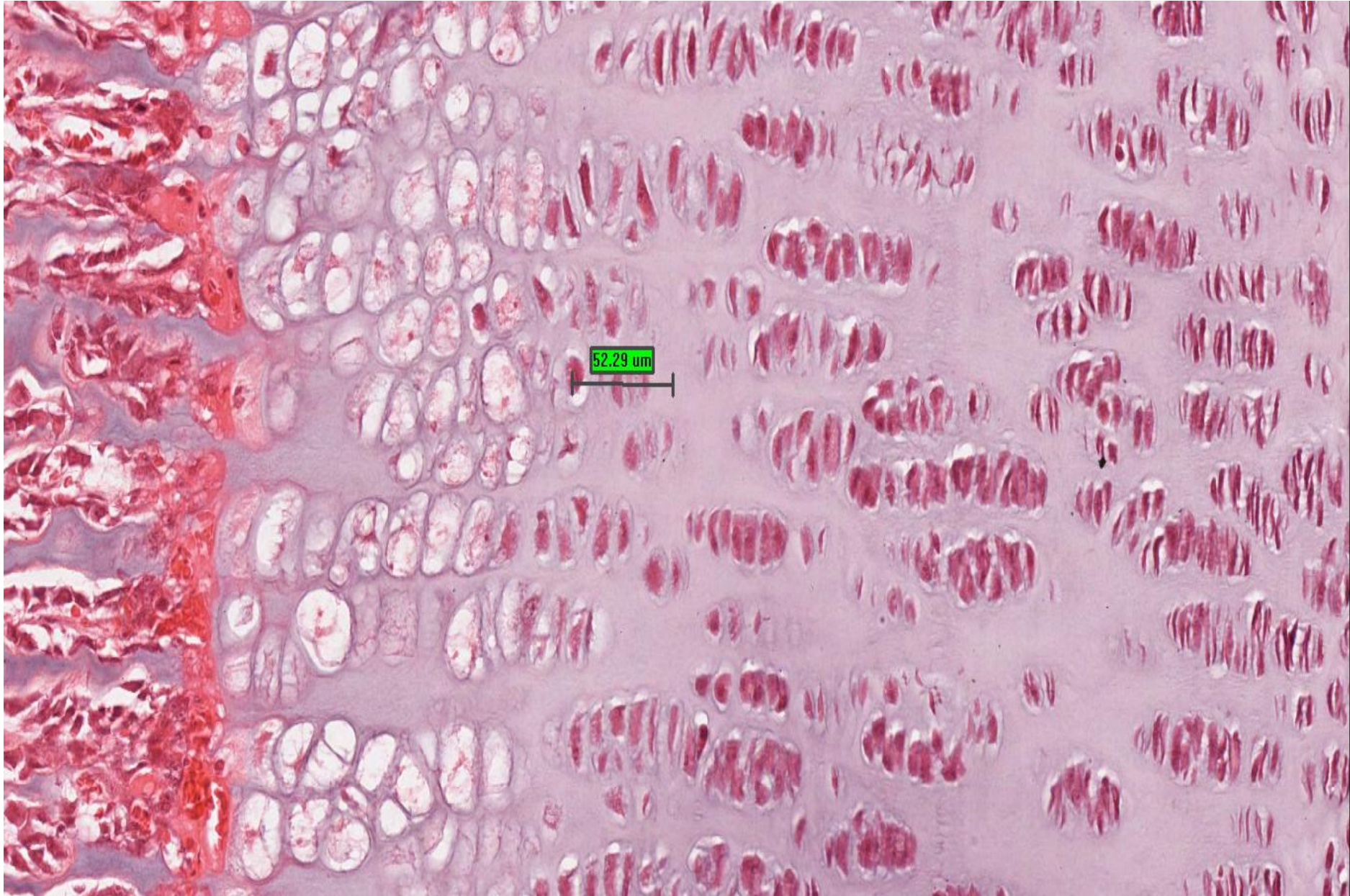


Epiphyseal Growth Plate/Disc





Zones of maturation and hypertrophy



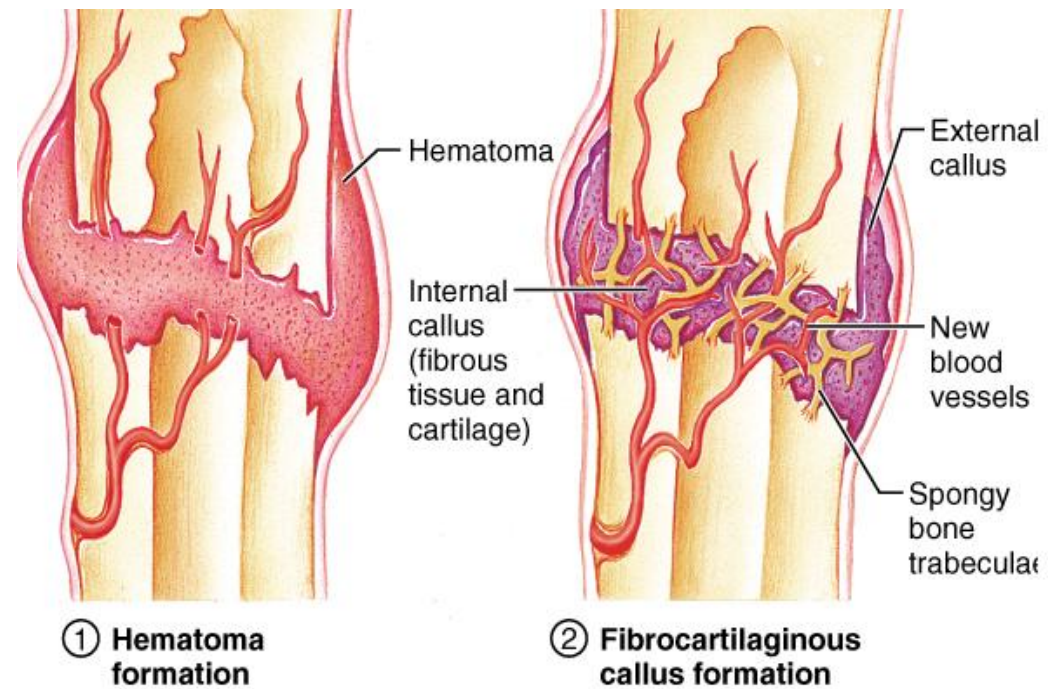
- At puberty, growth in bone length is increased dramatically by the combined activities of growth hormone, thyroid hormones, and the sex hormones.
- As a result osteoblasts begin producing bone faster than the rate of epiphyseal cartilage expansion.
- Thus the bone grows while the epiphyseal plate gets narrower and narrower and ultimately disappears.
- A remnant (**epiphyseal line**) is visible on X-rays or CT scan (*do you see them in the adjacent femur, tibia, and fibula?*)



Bone Fracture Repair

Step 1:

- A. Immediately after the fracture, extensive bleeding occurs. Over a period of several hours, a large blood clot, or **fracture hematoma**, develops.
- B. Bone cells at the site become deprived of nutrients and die. The site becomes swollen, painful, and inflamed.

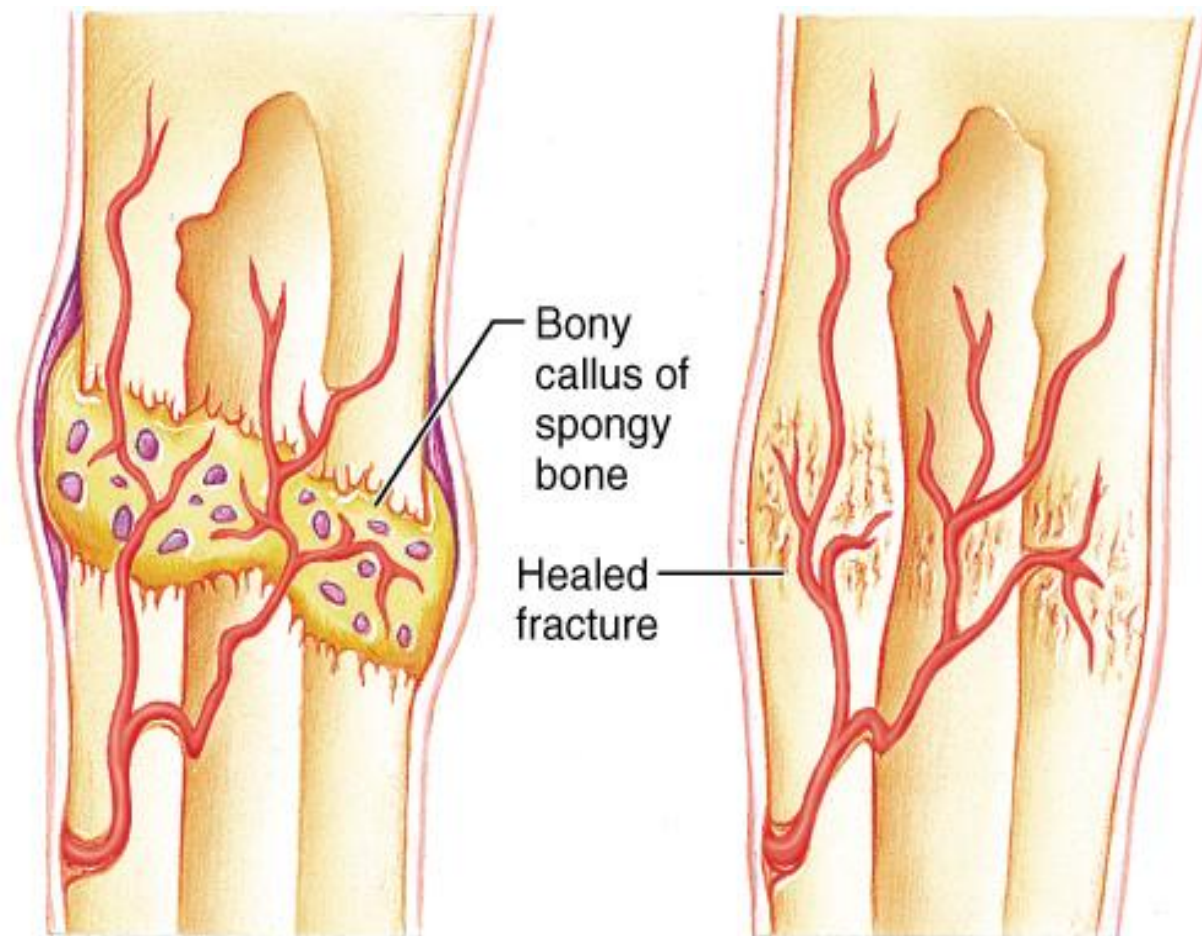


Step 2:

- A. Granulation tissue is formed as the hematoma is infiltrated by regenerating capillaries & **macrophages**, which begin to clean up the debris.
- B. Some **fibroblasts** produce collagen fibers that span the break, while others differentiate into chondroblasts which secrete cartilage matrix.
- C. Osteoblasts begin forming spongy bone.
- D. This entire structure is known as a **fibrocartilaginous callus** and it splints the broken bone.

Step 3:

- Bone trabeculae increase in number and convert the fibrocartilaginous callus into a **bony callus** of spongy bone (primary bone).
- Typically takes about 6-8 weeks for this to occur.

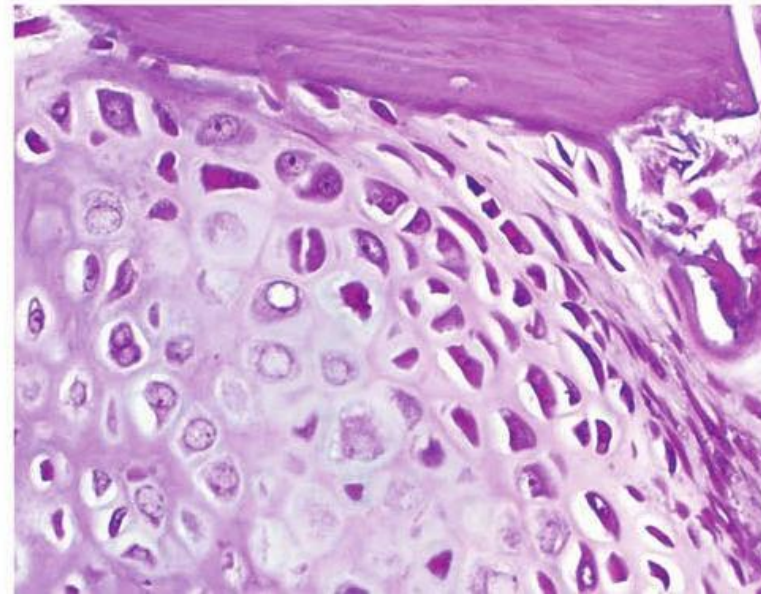
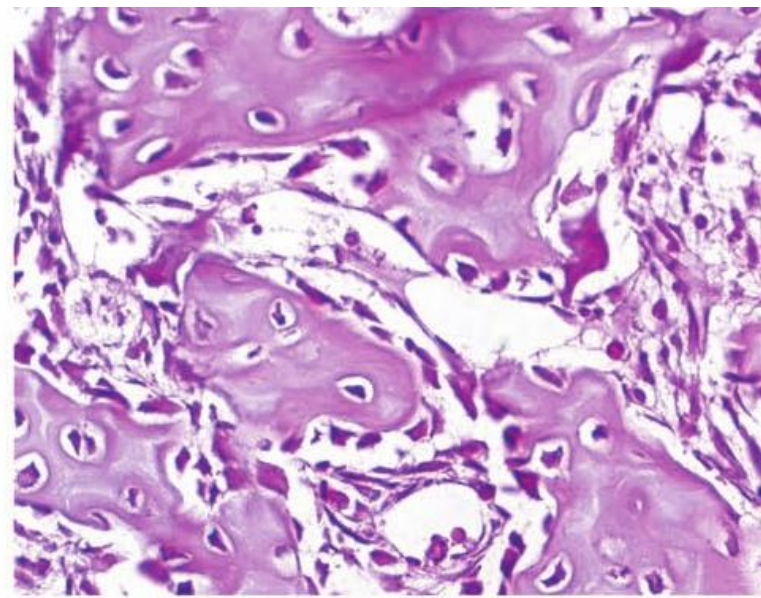
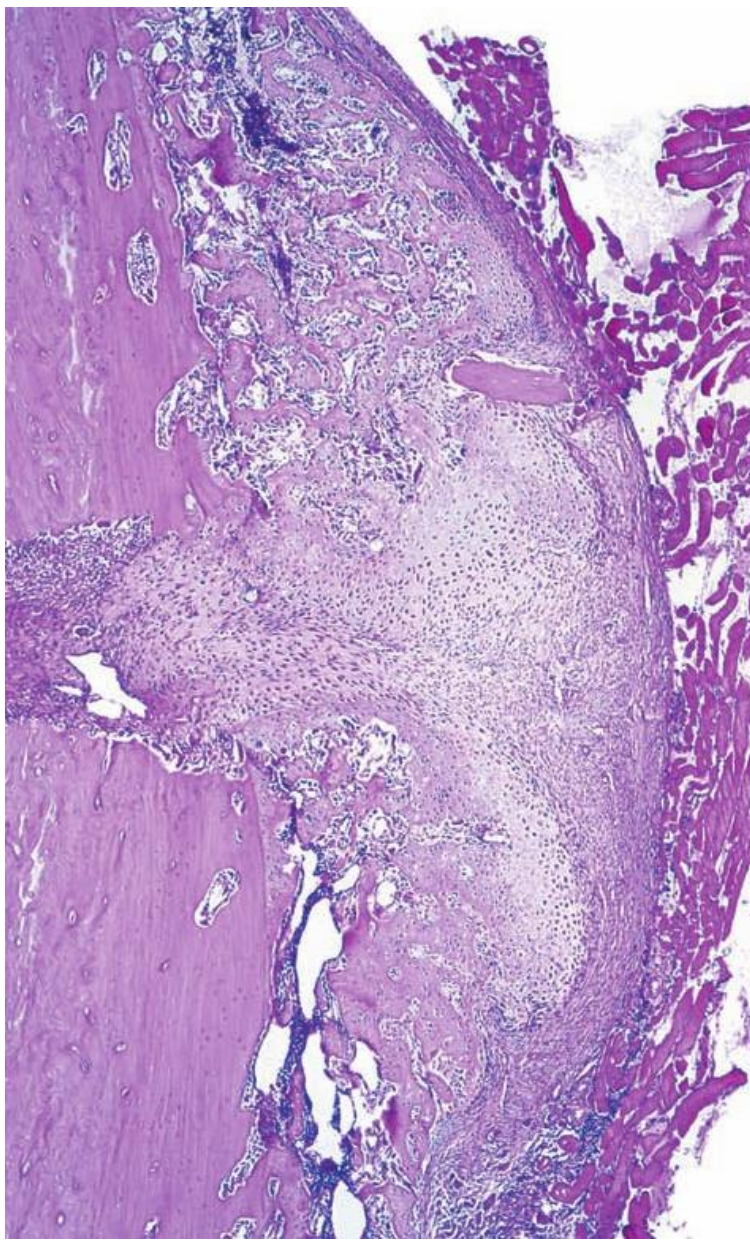


③ Bony callus formation

④ Bone remodeling

Step 4:

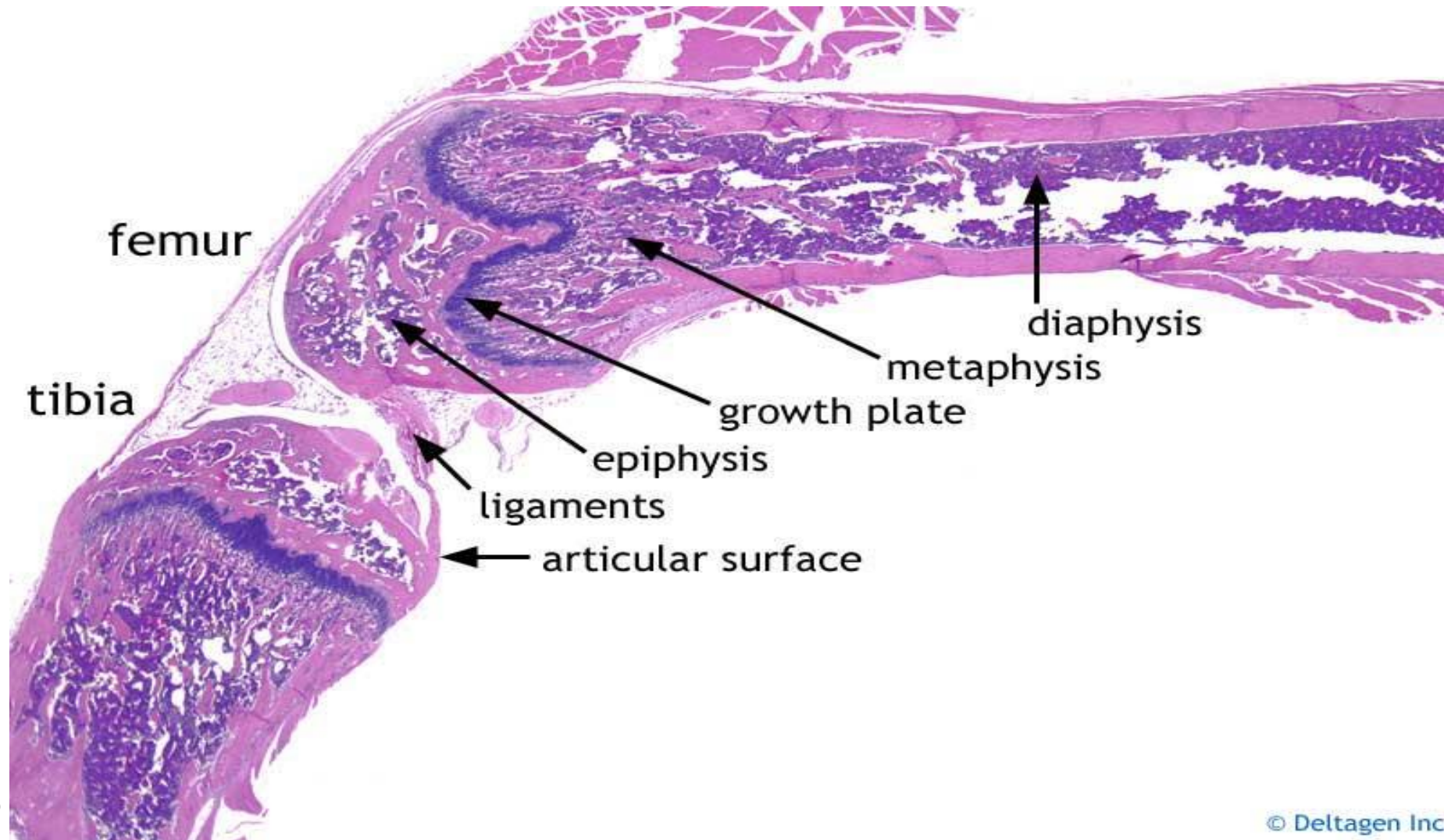
- During the next several months, the bony callus is continually remodeled.
- Osteoclasts work to remove the temporary supportive structures while osteoblasts rebuild the compact bone and reconstruct the bone so it returns to its original shape/structure.



Photomicrograph of fractured bone undergoing repair.

Joint supporting tissues

- Mostly dense collagenous connective tissues
- Regenerate fairly good but not as good as bone



SYNOVIAL JOINTS (DIARTHROSES)

- Typical synovial joint comprises:

1. **Articulating bones**

2. **Articular cartilage** (hyaline Cartilage that covers the ends of articulating bones) and often **articular disc(s)**.

3. **Ligaments** and a **articular capsule** of dense CT: maintain proper alignment of the bones.

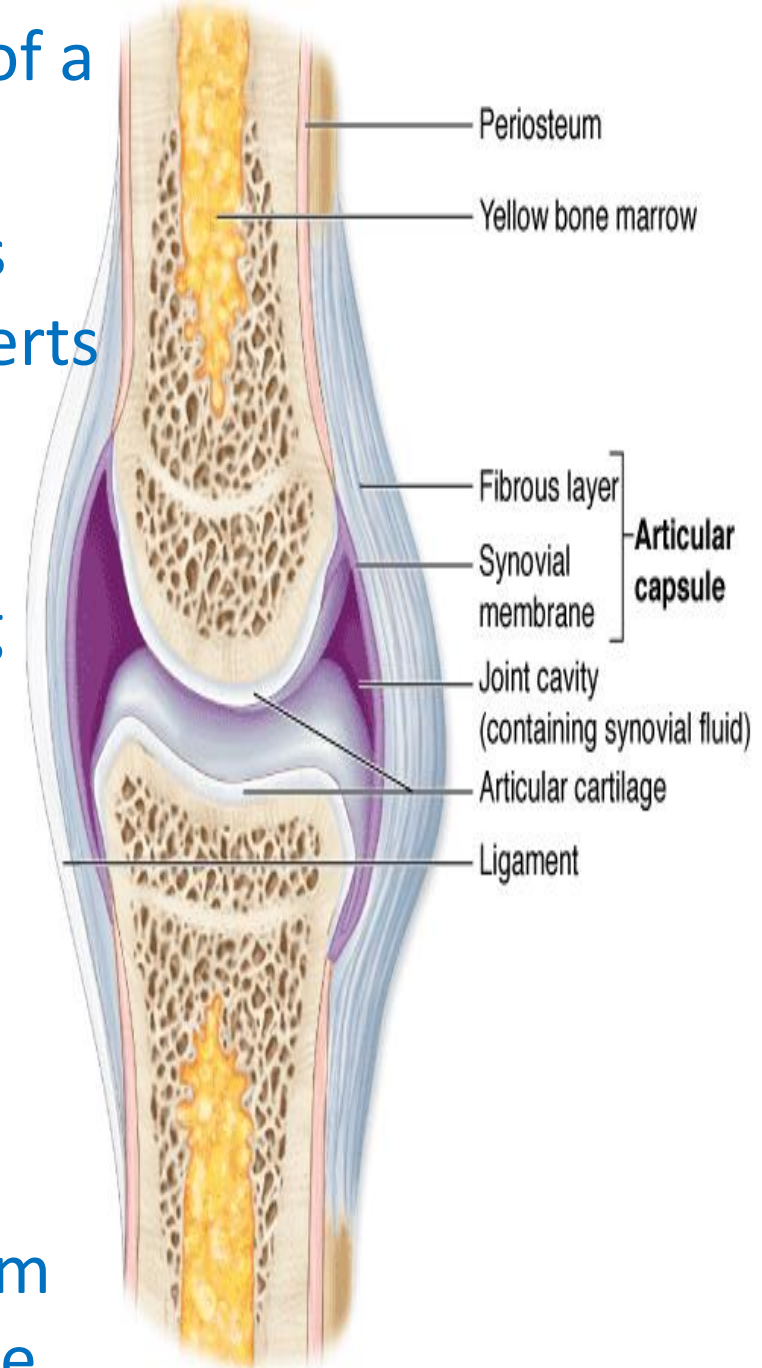
4. **The capsule encloses a sealed joint cavity** that contains **synovial fluid**, a colorless, transparent, viscous fluid.

5. **The joint cavity is not lined by epithelium, but by a specialized CT called the synovial membrane** which extends folds and villi into the cavity and secretes the lubricant **synovial fluid**.

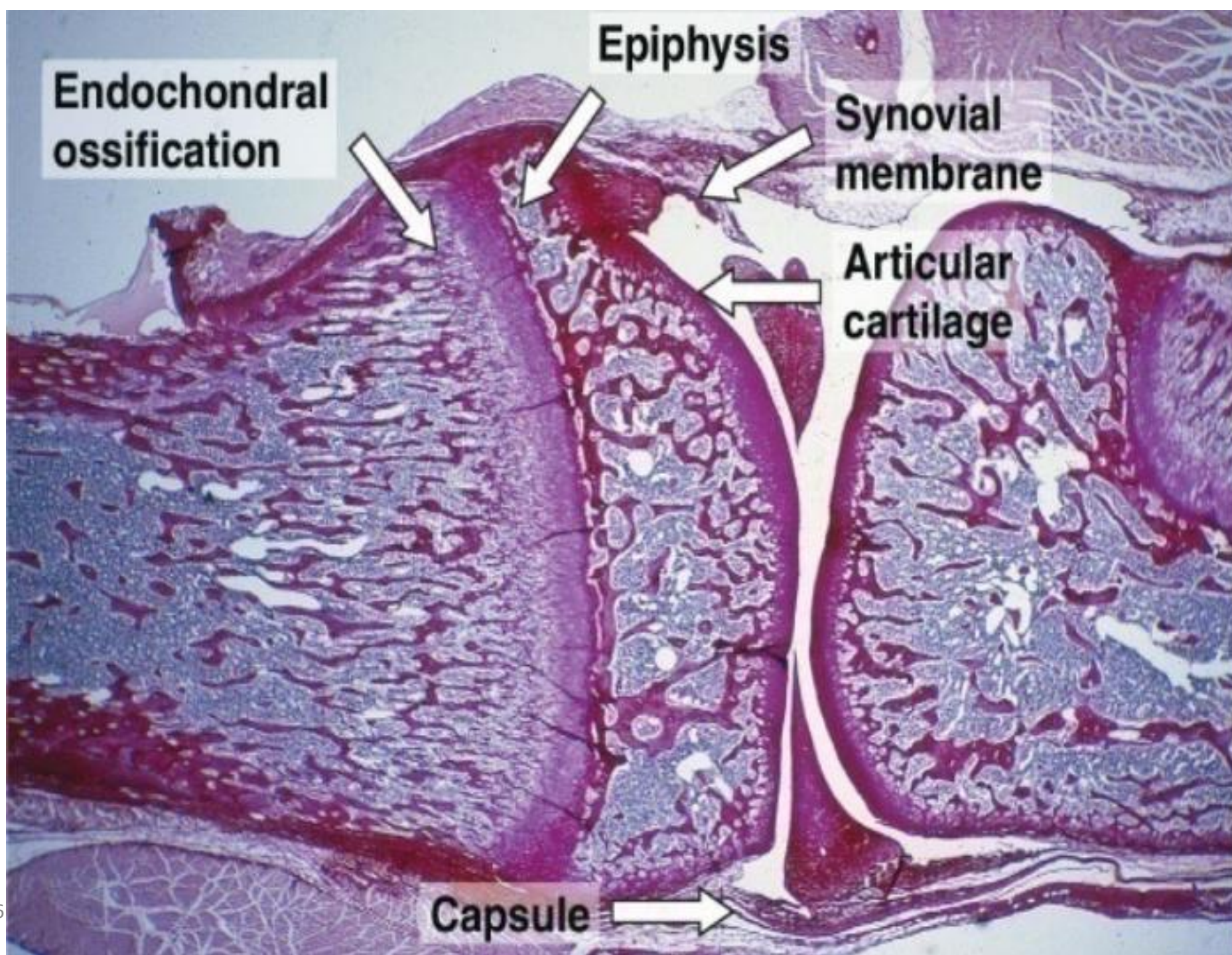
– **The synovial fluid is derived from blood plasma, but with a high concentration of hyaluronic acid produced by cells of the synovial membrane.**

This diagram shows the components of a synovial joint which include:

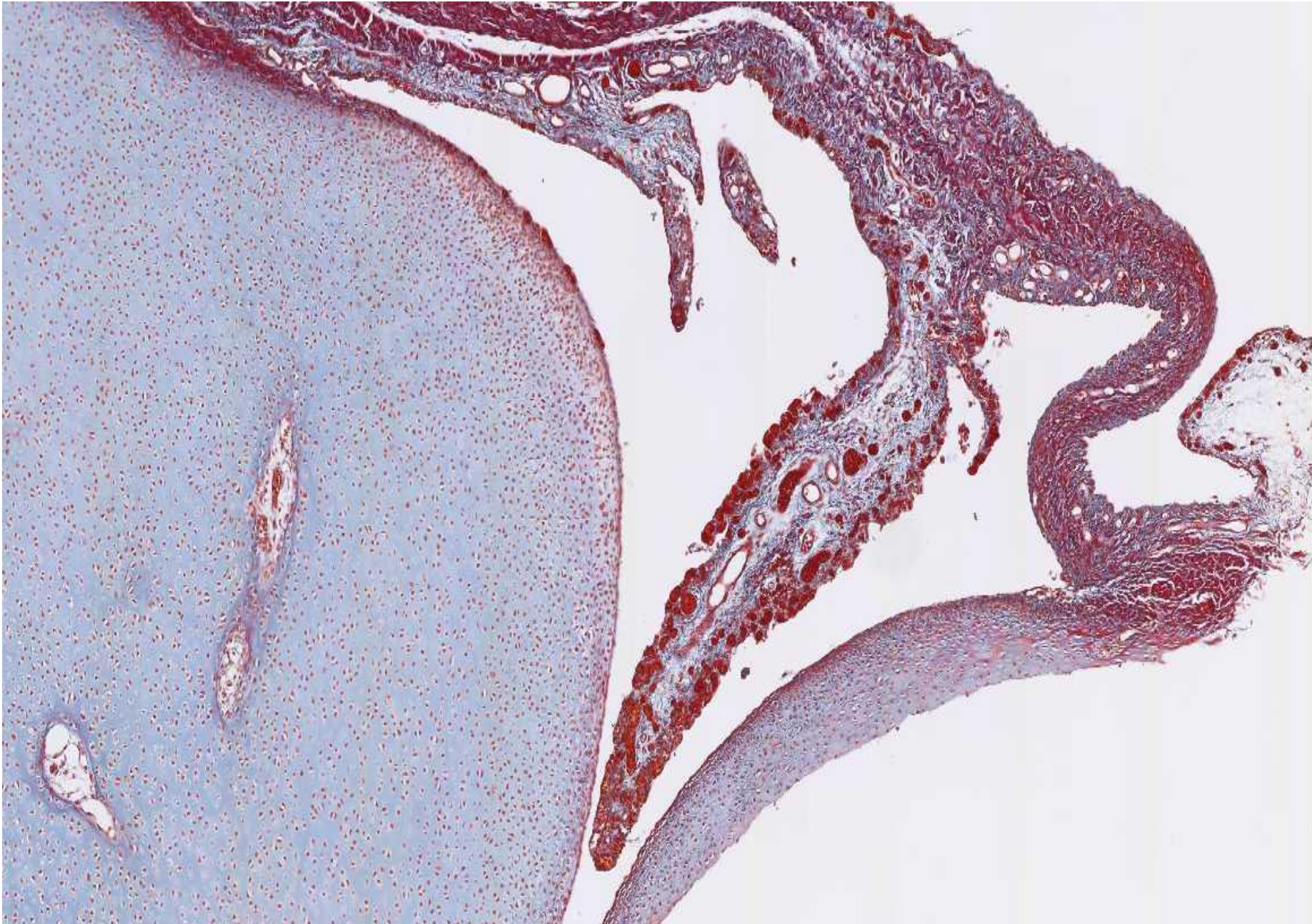
1. **a joint/articular capsule** continuous with a covering **ligament** which inserts into the periosteum of both bones,
 2. **a synovial or joint cavity** lined by **synovial membrane** and containing synovial fluid as a lubricant, and
 3. **the ends of articulating bones, epiphyses** covered by
 4. **an articular cartilage.**
- Also shown here are the folds of synovial membrane which extend prominently into the joint cavity from connective tissue of the joint capsule for production of synovial fluid



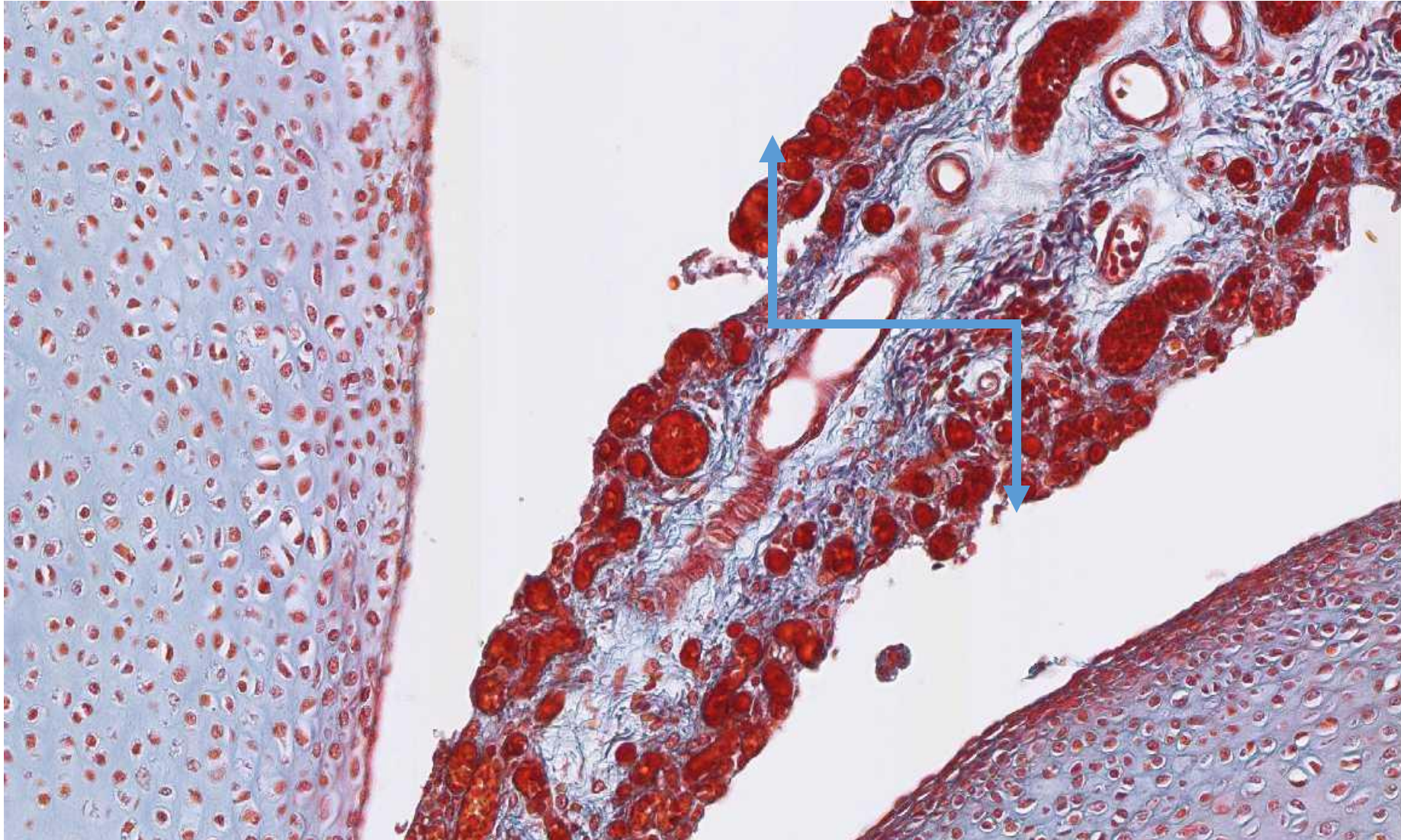
a Typical synovial joint



Key Components a Synovial Joint



- Depending on the location, the bulk of the synovial tissue may be of loose collagenous type (areolar synovium), of more dense collagenous type (fibrous synovium), or predominantly composed of fat (adipose synovium) as in the case of intra-articular fat pads.



Macrophage
like cell

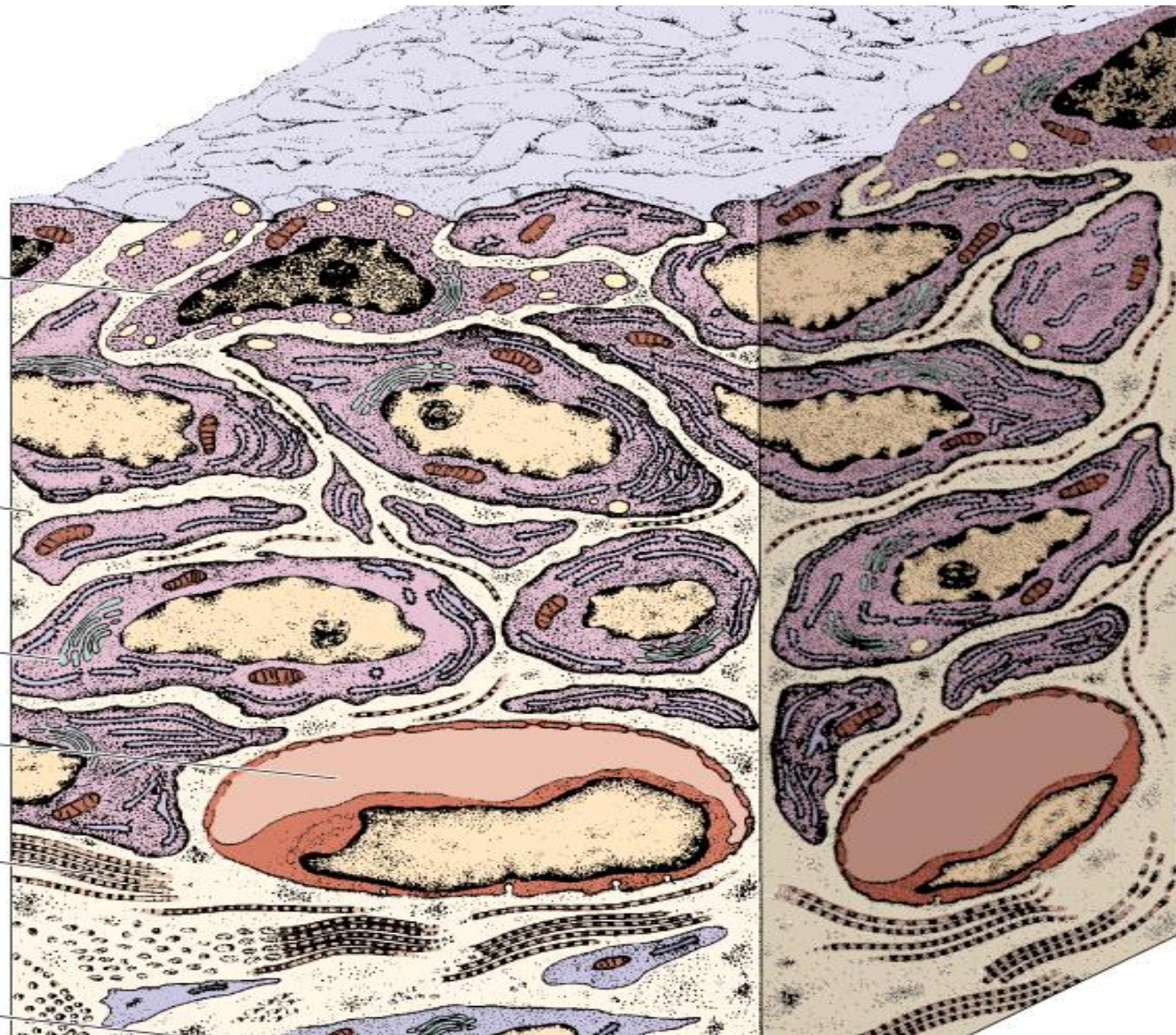
Ground
substance

Fibroblastlike
cell

Fenestrated
blood capillary

Collagen
fibrils

Fibroblast

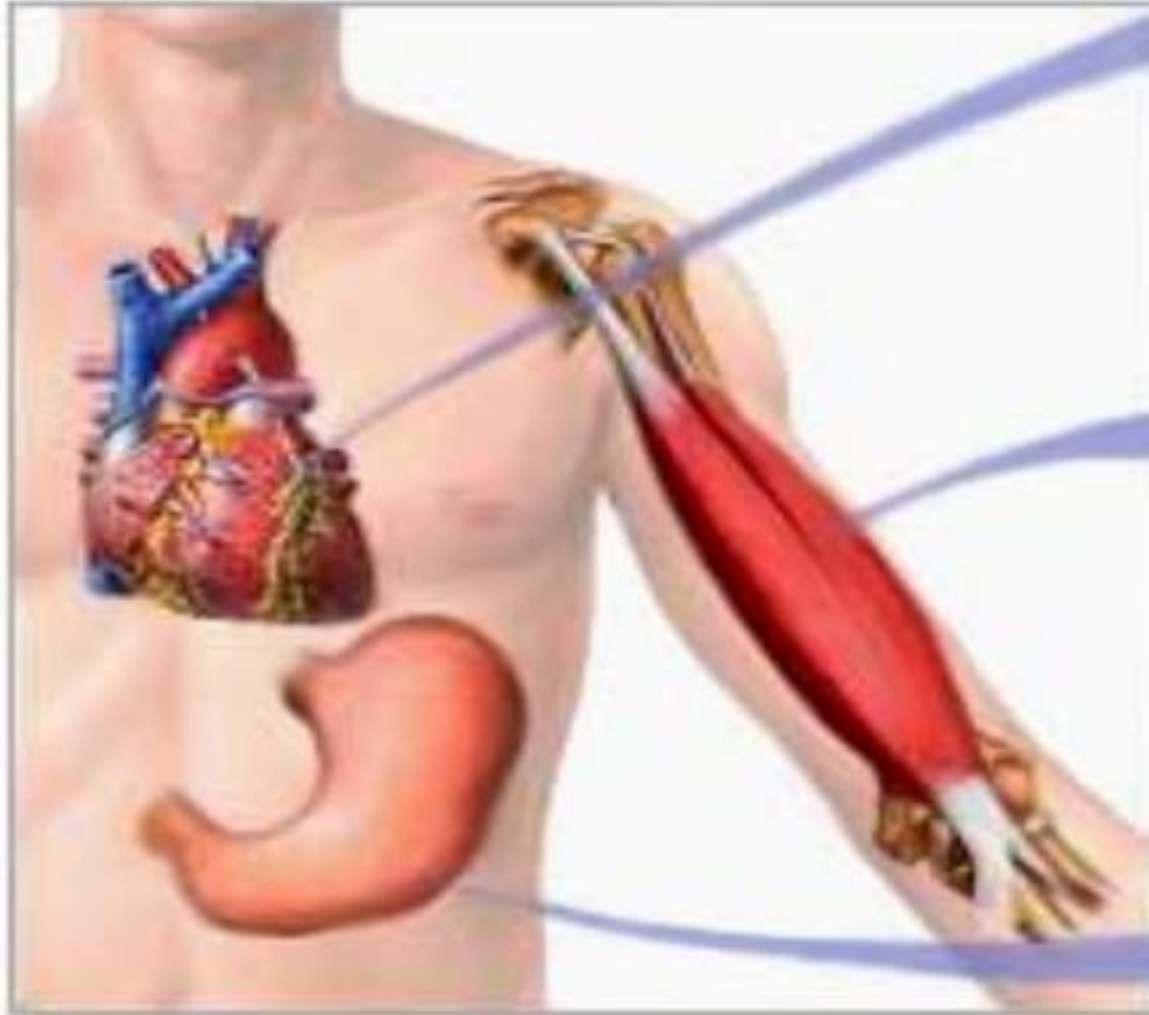


MUSCLE TISSUES

- **Muscle tissue** is one of the four basic tissue types.
- Muscle tissue combined with various connective tissues, nerves, and blood vessels makes up **muscle organs** that are familiar to us.
- Muscle organs are quite complex and are a wonder of both biology and physics.



THREE BASIC TYPES OF MUSCLE TISSUES



Cardiac muscle cell

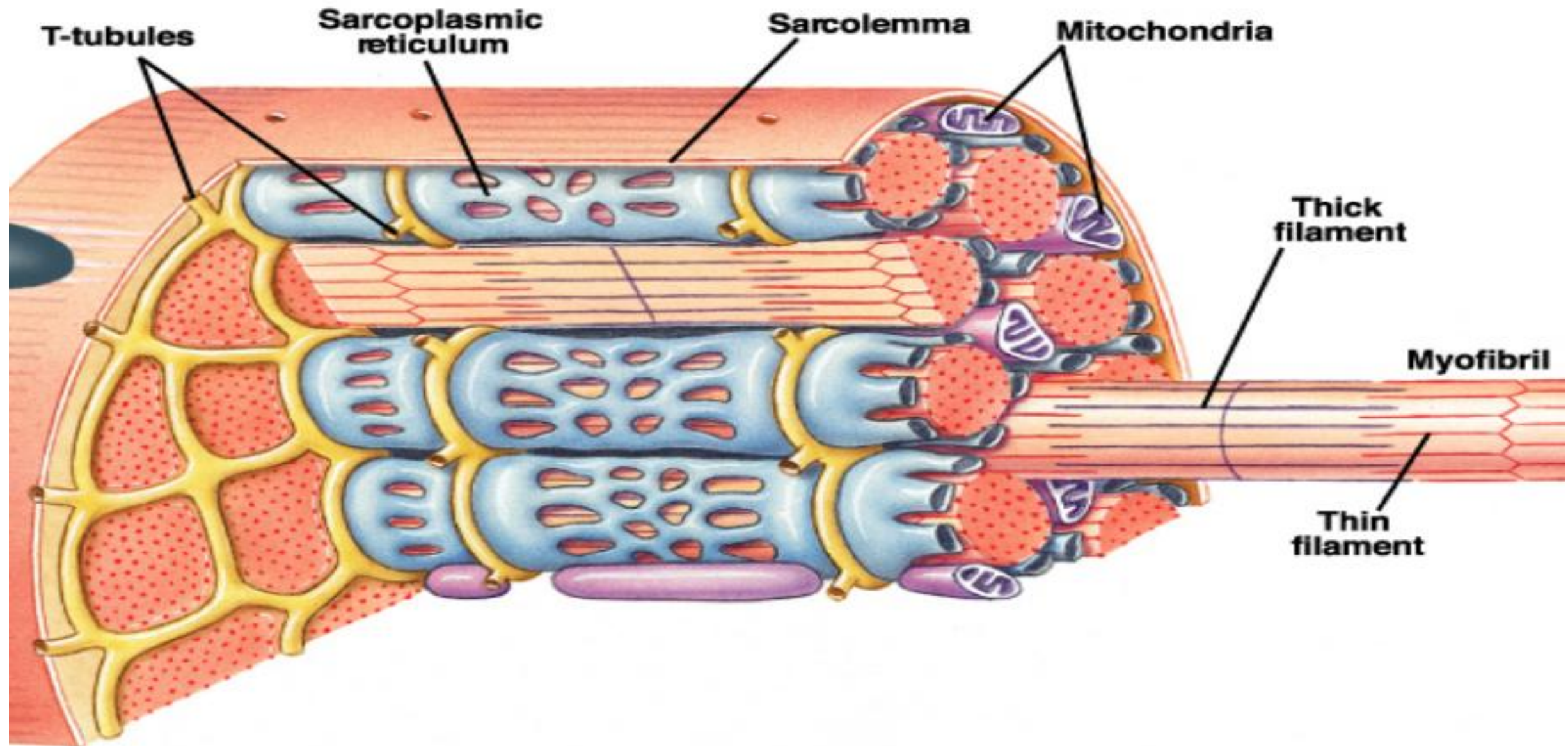


Skeletal muscle cell



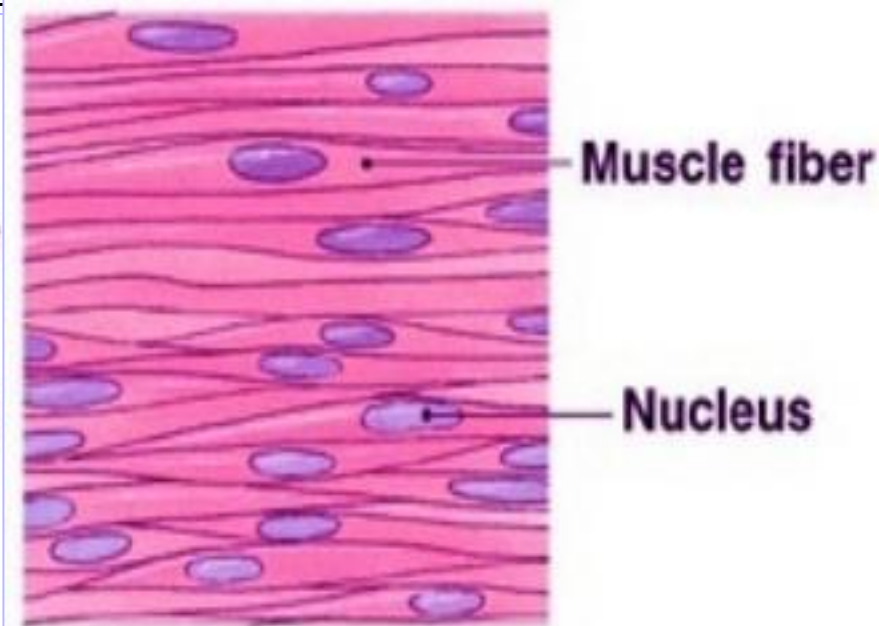
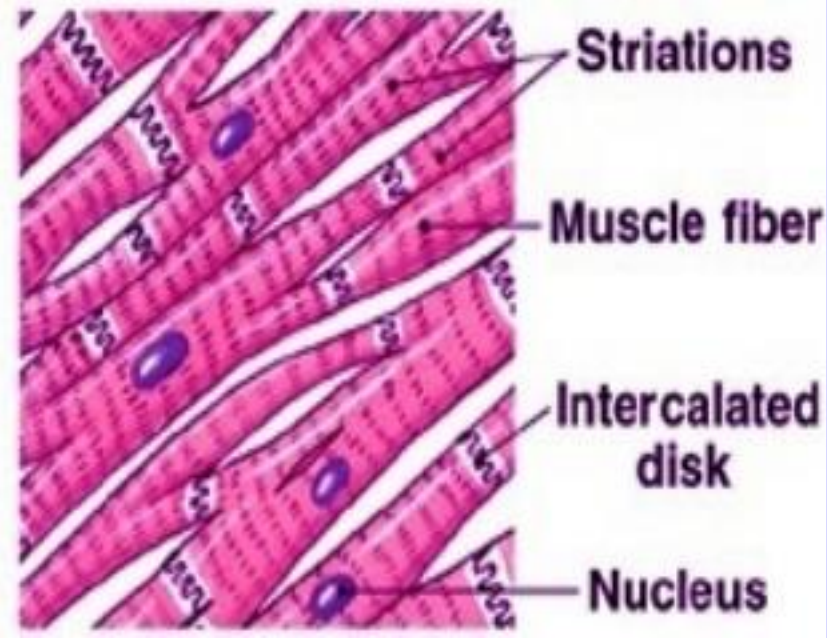
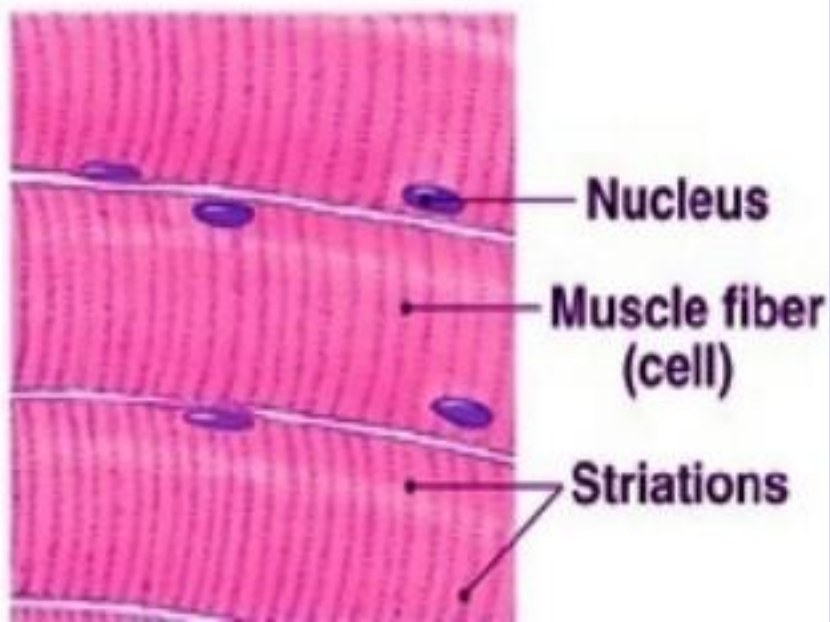
Smooth muscle cell

Cellular Components of a striated muscle fiber

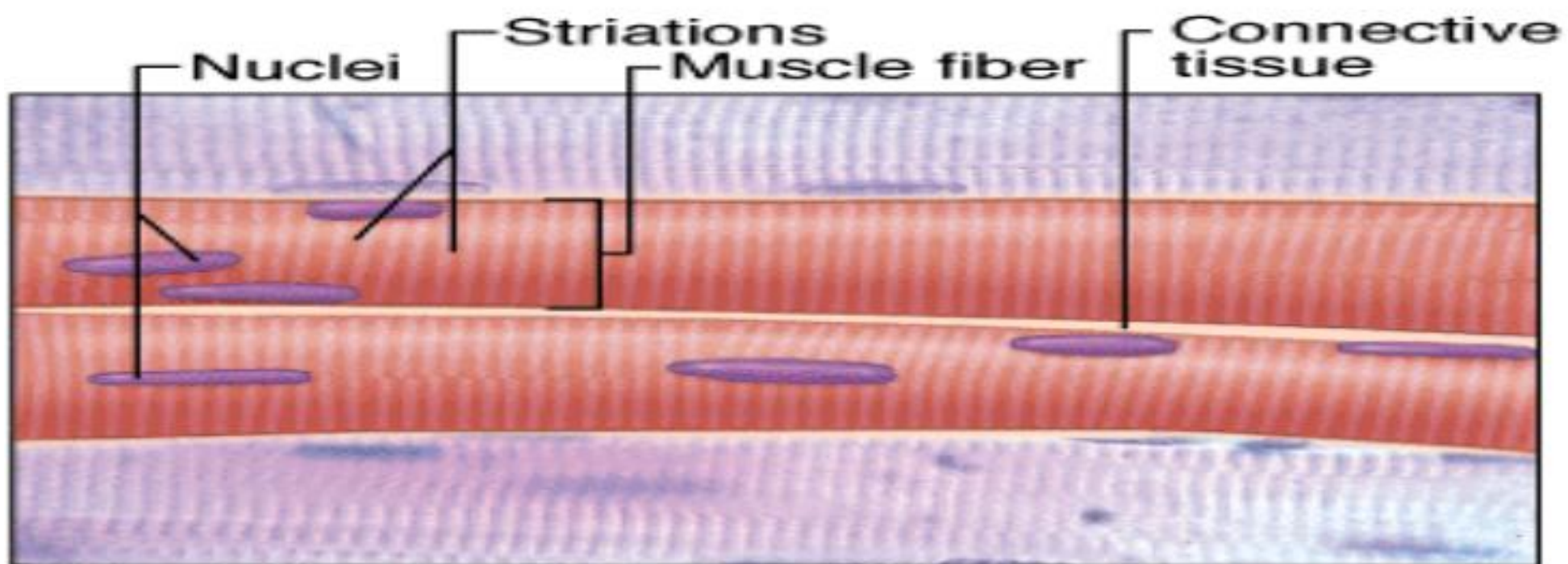
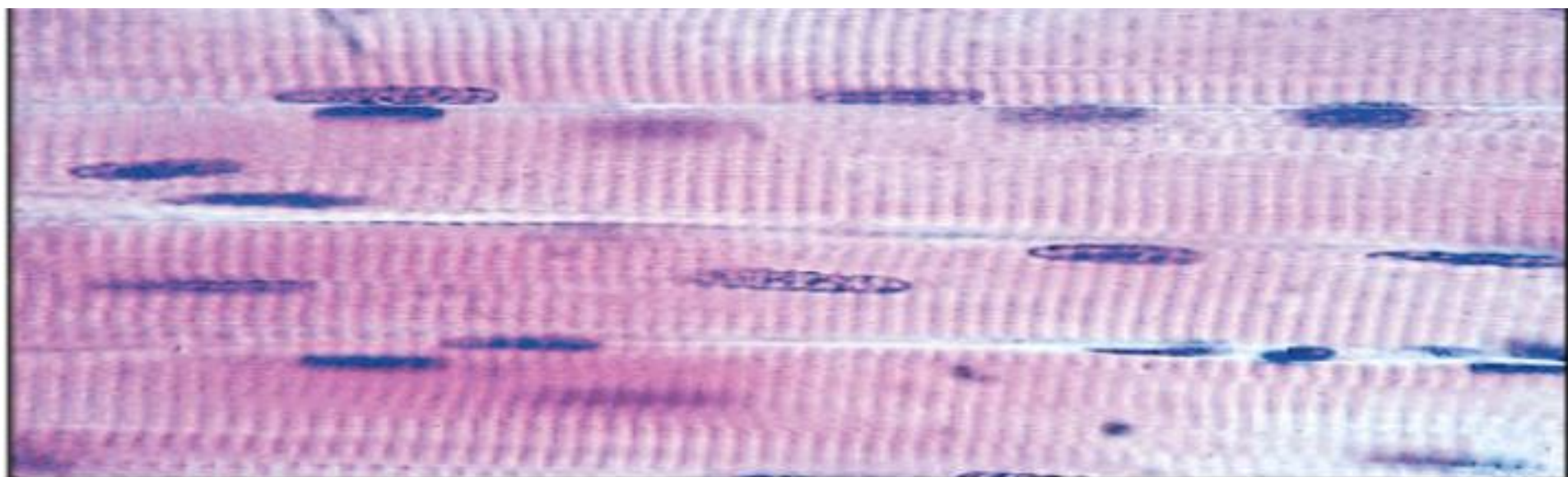


Muscle Tissues

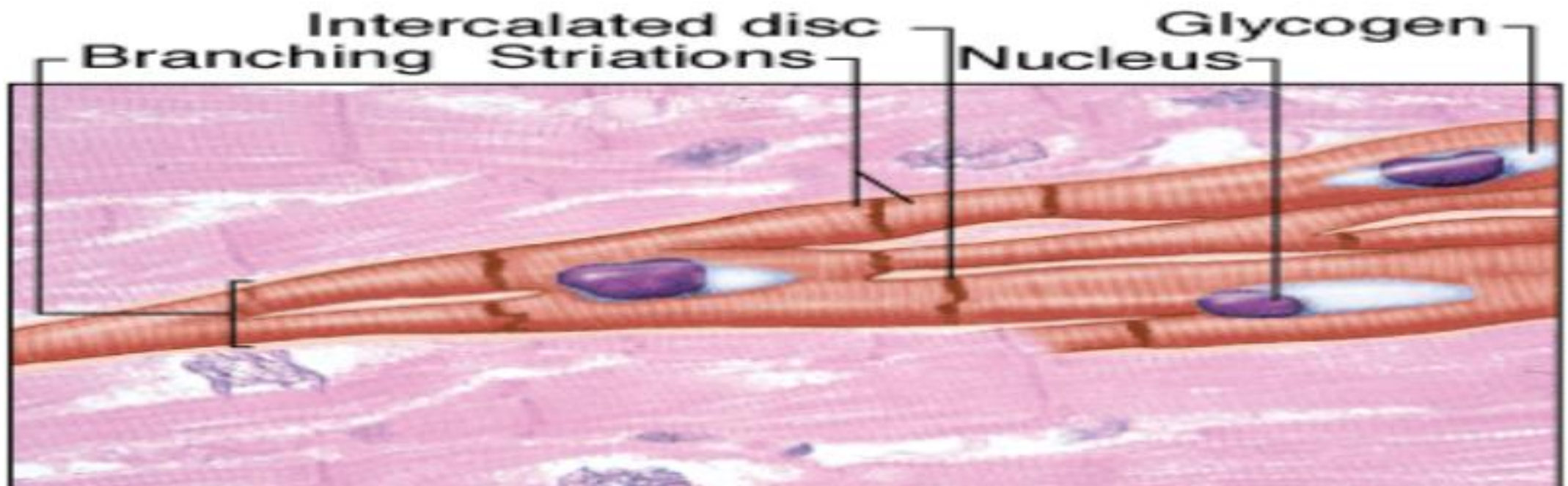
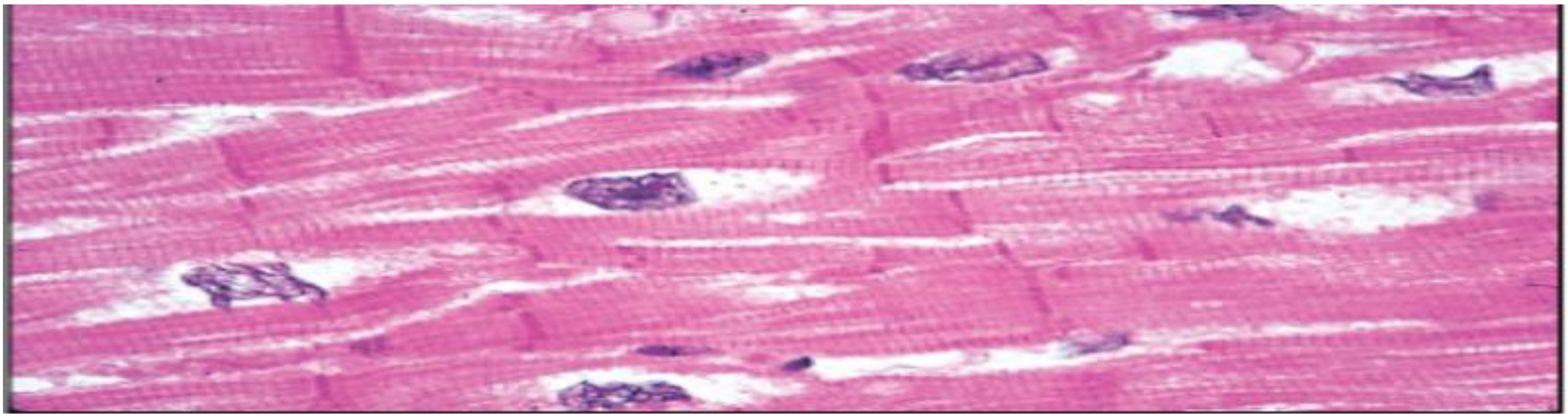
Skeletal muscle	Cardiac muscle	Smooth muscle
Location: attached to the skeleton, in soft palate, lip, tongue, pharynx, esophagus	Found in the myocardium of the heart & its great vessels	Found in the blood vessels & viscera
Striated	Striated	Non-striated
Voluntary	Involuntary	Involuntary



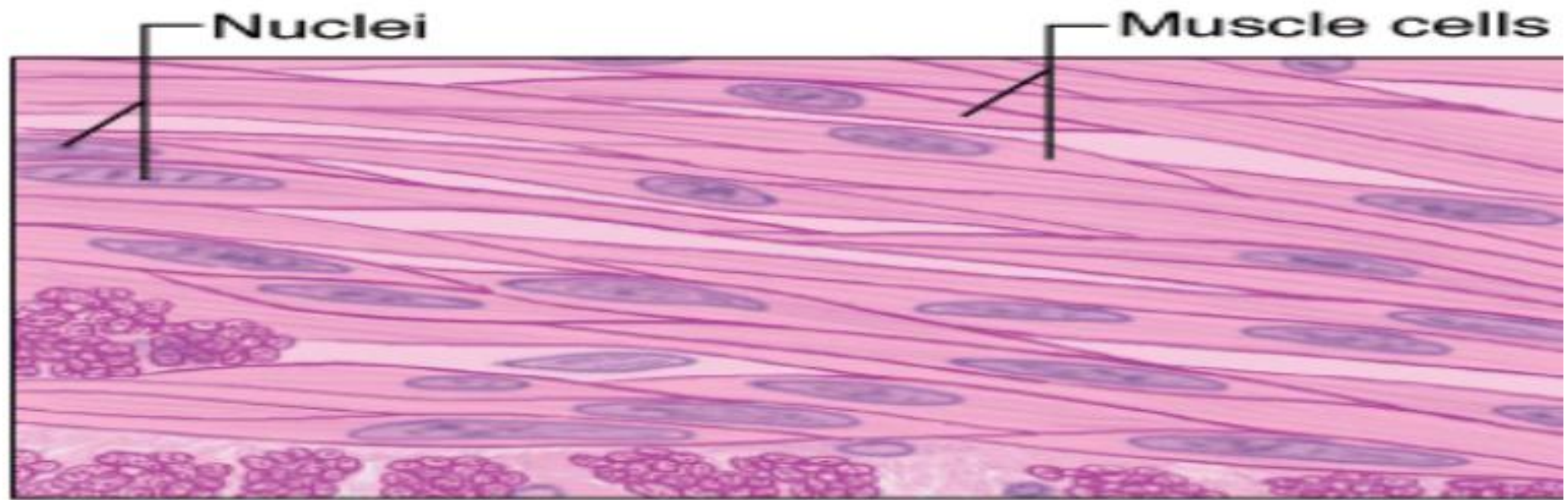
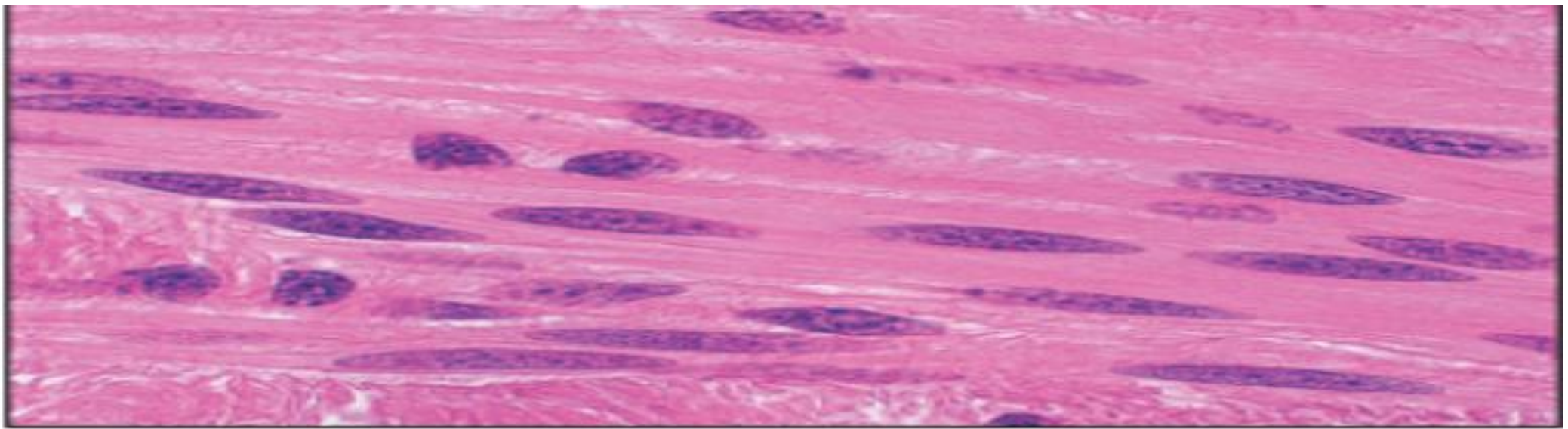
Type	Skeletal muscle	Cardiac muscle	Smooth muscle
Location	skeleton, tongue, lip, eyelid, palate, ear esophagus, pharynx, larynx, anal canal	heart wall, SVC, IVC, pulmonary veins	visceral organs, vessels, dermis of skin, middle layer of eyeball, trachea bronchi, bronchioles, GIT, genital ducts, urinary tract
CT Sheath	epi-, peri-, & endo-	Endo- & peri	Endo- & perimysium
Cell shape	long cylindrical, non-branching fibers	Cylindrical, branching & anastomosing cells	small fusiform or spindle-shaped cells
Number & location of nuclei	Many nuclei up to 100 per cell, peripherally located	One or two nuclei, centrally located	one per cell, centrally located
Cross-striations	present	present	absent
Intercalated Ds	absent	present	absent
T-system	T-Triad	T-Diad	Absent, instead endomembrane system
Control	voluntary	involuntary	involuntary
Growth and regeneration response	Hypertrophy, and limited hyperplasia	No hyperplasia, only hypertrophy	Unlimited, both Hypertrophy and hyperplasia



a Skeletal muscle



b. Cardiac muscle tissues



c Smooth muscle